

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	7
MDMT Summary:	8
Warnings and Errors:	10
Input Echo:	11
XY Coordinate Calculations:	20
Internal Pressure Calculations:	21
Element: Bottom head	22
Element: Shell	24
Element: Top head	26
Shop Pressure Test Results:	28
Field Pressure Test Results:	30
External Pressure Calculations:	32
Element and Detail Weights:	33
Nozzle Flange MAWP:	36
Natural Frequency Calculation:	37
Forces/Moments Applied to Vessel	38
Wind Load Calculation:	39
Earthquake Load Calculation:	42
User Force/Moment Shear and Bendi	44
Wind/Earthquake Shear, Bending:	45
Wind/Earthquake Deflection:	46
Longitudinal Stress Constants:	47
Longitudinal Allowable Stresses:	48
Longitudinal Stresses due to:	49
Stress due to Combined Loads:	51
Skirt Stress Calculations:	55
Skirt Stress Calculations:	59
Center of Gravity Calculation:	66
Basering Calculations:	67
Nozzle Calcs.: N2_400	71
Nozzle Calcs.: U1_600	81
Nozzle Calcs.: K4_80	87
Nozzle Calcs.: K1_50	96
Nozzle Calcs.: K6_50	99
Nozzle Calcs.: K3_50	102
Nozzle Calcs.: M1_600	105
Nozzle Calcs.: K2_50	111
Nozzle Calcs.: K5_80	118
Nozzle Calcs.: N1_400	127
Problems/Failures Summary:	137



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\108-DA-Rev 01.pvdb

Date of Analysis : Sep 9, 2026 5:12pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 3660.000 mm ID	
Vessel Design Length, Tangent to Tangent	2975.00 mm
Distance of Bottom Tangent above Grade	3800.00 mm
Distance of Base above Grade	0.00 mm
Specified Datum Line Distance	3800.00 mm
Internal Design Temperature	400 °C
Internal Design Pressure	6.416 MPa
External Design Temperature	400 °C
External Design Pressure	0.103 MPa
Maximum Allowable Working Pressure	6 MPa
External Max. Allowable Working Pressure	2 MPa
Shop Test Pressure	11.735 MPa
Field Test Pressure	11.069 MPa
Required Minimum Design Metal Temperature	-12.0 °C
Warmest Computed Minimum Design Metal Temperature	7.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code	ASCE/SEI 7-16
Earthquake Design Code	ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+IT
Head	SA-516 70	...	...	K02700	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-516 70	...	...	K02700	+IT
Nozzle	SA-105	...	...	K03504	+IT
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Basering	SA-36	...	...	K02600	
Base Bolting	SA-307 A	...	6 <= t <= 100		

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-2	...	...	...	1.6000	N/A	No
Skirt-1	...	...	...	1.6000	N/A	Yes
Bottom head	6.416	0.10	6.000	3.0000	N/A	Yes
Shell	6.416	0.10	6.000	3.0000	N/A	Yes
Top head	6.416	0.10	6.000	3.0000	N/A	Yes

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
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Skirt	-1000.0	2800.0	32.0	32.0	...	...	0.50	0.50
Skirt	0.0	1000.0	32.0	32.0	...	...	0.70	0.50
Sphere	0.0	0.0	70.0	63.0	62.6	9.1	1.00	1.00
Cylinder	2975.0	2975.0	125.0	125.0	124.0	17.0	1.00	1.00
Sphere	2975.0	0.0	70.0	63.0	62.6	9.1	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N2_400	WRC-107/537	0.987	n/a	Passed
N1_400	WRC-107/537	0.987	n/a	Passed

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1400	24078.3	223549	126776744	1256183680
20	30	3300	19278.7	218769	66052336	636687360
30	40	3799.98	17564.5	215485	47623784	419478720
40	50	5287.5	17564.5	192011	47623244	419472416
50	60	6774.98	9070.67	56461.8	7987661	49720448

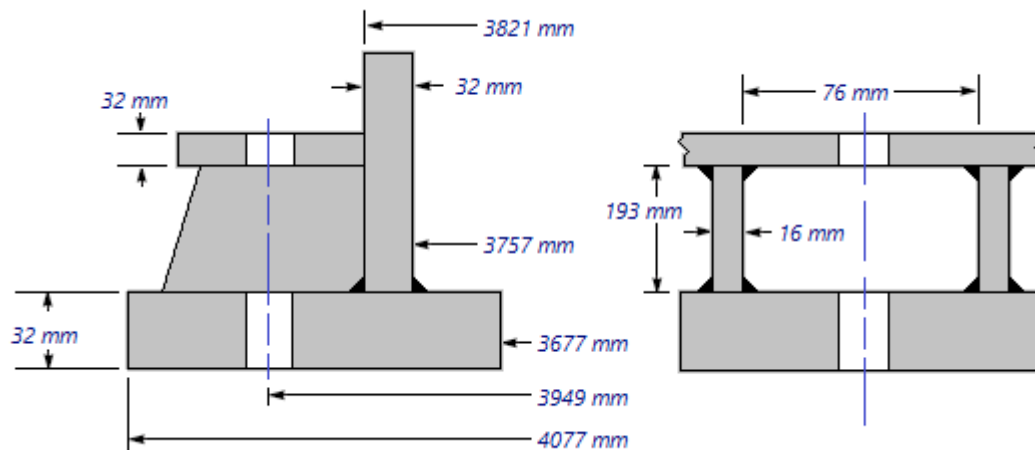
Abs Max of the all of the Stress Ratio's : 0.9712

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing 32.0000 mm
 Inside Diameter of Basing 3677.0000 mm
 Outside Diameter of Basing 4077.0000 mm
 Nominal Diameter of Bolts 36.0000 mm
 Diameter of Bolt Circle 3949.0000 mm
 Number of Bolts 24

Thickness of Gusset Plates 16.0000 mm
 Average Width of Gusset Plates 118.0000 mm
 Height of Gussets 193.0000 mm
 Distance between Gussets 76.0000 mm
 Thickness of Top Plate or Ring 32.0000 mm
 Radial Width of the Top Plate 128.0000 mm
 Number of Gussets 48

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View



Loads for Foundation/Support Design:

Total Wind Shear on Support	54399.	N
Total Earthquake Shear on Support	253870.	N
Wind Moment on Support	415945472.	N-mm
Earthquake Moment on Support	1545352448.	N-mm
Maximum Initial Bolt Load	6417.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	92731.8	kgm
Shop Test - Fabricated + water (full)	151204.0	kgm
Shipping - Fab. + rem. intls.+ shipping app.	136640.0	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	137340.0	kgm
Empty - Fab. + intls. + details + wghts.	137340.0	kgm
Operating - Empty + operating liquid (no ca)	145390.0	kgm
Field Test - Empty weight + test medium (full)	162122.5	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N2_400	6	OK	...	OK	17.25	OK	Passed
U1_600	6	OK	...	OK	21.59	OK	Passed
K4_80	6	OK	...	OK	8.26	OK	Passed
K1_50	6	...	...	OK	7.80	OK	No Calc[*]
K6_50	6	...	...	OK	7.80	OK	No Calc[*]
K3_50	6	...	...	OK	7.80	OK	No Calc[*]
M1_600	6	OK	...	...	...	OK	Passed
K2_50	6	OK	...	OK	7.80	OK	Passed
K2_50	6	...	...	OK	7.80	OK	No Calc[*]
K5_80	6	OK	...	OK	7.80	OK	Passed
N1_400	6	OK	...	OK	17.25	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 6 Nozzle : K480
 Minimum MAWP Shells/Flanges : 6 Element : Top head
 Minimum MAPnc Shells/Flanges : 9 Element : Shell

Computed Vessel M.A.W.P. : 6 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N2_400	0.000	0.000	739.151
30	U1_600	0.000	180.000	1124.000
30	K4_80	0.000	0.000	228.000
40	K1_50	100.000	0.000	328.328
40	K6_50	2000.001	0.000	328.328
40	K3_50	2500.000	180.000	328.328
50	M1_600	0.000	0.000	1124.000
50	K2_50	0.000	0.000	203.600
50	K5_80	0.000	180.000	208.900
50	N1_400	0.000	0.000	739.151

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

Diameter Limit clash between : N1400 and K250.
 The Centerline Dist. is 0.00 while the sum of the limit Radii is 471.38 mm.

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
K2_50	50 mm	Actual	WNF	83.6	16.800	184.33	74.00	962	600
K1_50	50 mm	Actual	LWN	84.3	16.764	...	...	624	600
K6_50	50 mm	Actual	LWN	84.3	16.764	...	...	624	600
K3_50	50 mm	Actual	LWN	84.3	16.764	...	...	425	600
K4_80	80 mm	Actual	WNF	108.0	14.000	244.00	62.00	371	600
K5_80	80 mm	160	WNF	88.9	11.125	288.90	74.00	1366	600
N2_400	400 mm	80	WNF	406.4	21.412	723.85	146.00	563	600
N1_400	400 mm	80	WNF	406.4	21.412	723.85	146.00	563	600
U1_600	600 mm	Actual	WNF	600.0	22.000	1120.00	62.00	2563	600
M1_600	600 mm	Actual	WNF	600.0	22.000	1200.00	64.00	537	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K2_50	SA-106 B	62.000	10.000	56.000	74.000	...
K1_50	SA-105	124.000	10.000	...	...	...
K6_50	SA-105	124.000	10.000	...	...	...
K3_50	SA-105	124.000	10.000	...	...	...
K4_80	SA-106 B	62.000	10.000	48.000	62.000	...
K5_80	SA-106 B	62.000	10.000	56.000	74.000	...
N2_400	SA-106 B	62.000	10.000	110.000	146.000	14.000
N1_400	SA-106 B	62.000	10.000	110.000	146.000	14.000
U1_600	SA-516 70	62.000	14.000	62.000	62.000	14.000
M1_600	SA-516 70	62.000	20.000	56.000	60.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K2_50	...	0.0	800.00	0.00	Top head
K1_50	100.001	0.0	300.00	200.00	Shell
K6_50	2000.001	0.0	300.00	200.00	Shell
K3_50	2500.000	180.0	300.00	0.00	Shell
K4_80	...	0.0	190.50	0.00	Bottom head
K5_80	...	180.0	1200.00	0.00	Top head
N2_400	...	0.0	200.00	300.00	Bottom head
N1_400	...	0.0	200.00	300.00	Top head
U1_600	...	180.0	1500.00	1000.00	Bottom head
M1_600	...	0.0	450.00	0.00	Top head



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	SPHERICAL HEAD: 70.0mm THK X 3722.0mm ID X 0.0mm	SA-516 70
1	INSULATION: 1825mm X 150mm THK	
1	PLATFORM: 1000mm X 1500mm WIDE	
1	LADDER: 4300.0mm	...
1	PACKING: 2975.000mm	...
1	INSULATION: 2975mm X 150mm THK	
1	PLATFORM: 1000mm X 4000mm WIDE	
1	LADDER: 5200.0mm	...
1	INSULATION: 1861mm X 150mm THK	
2	CLASS 600 GR 1.1, 600.0mm BLIND FLANGE(S)	SA-105
2	CLASS 600 GR 1.1, 80.0mm BLIND FLANGE(S)	SA-105
3	CLASS 600 GR 1.1, 50.8mm BLIND FLANGE(S)	SA-105
1	CLASS 600 GR 1.1, 50.0mm BLIND FLANGE(S)	SA-105
1	BASERING: 4077.0mm OD X 3677.0 ID X 32.0mm	SA-36
24	BASERING BOLTS: 36.0mm DIA	SA-307 A
24	BASERING BOLT NUTS: 36.0mm DIA	...
1	BASERING TOP RING: 4077.0mm OD X 3821.0 ID X 32.0mm	...
48	BASERING GUSSET PLATES: 118.0mm X 118.0mm X 193.0mm X 16.0mm	...
3	CYLINDRICAL SKIRT SEGMENT 32.0mm THK X 2622.8mm X 2800.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 32.0mm THK X 2655.6mm X 2800.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 32.0mm THK X 1278.6mm X 2800.0mm	SA-516 70
3	CYLINDRICAL SKIRT SEGMENT 32.0mm THK X 2622.8mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 32.0mm THK X 2655.6mm X 1000.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 32.0mm THK X 1278.6mm X 1000.0mm	SA-516 70
2	CYLINDRICAL SEGMENT 125.0mm THK X 4567.3mm X 2975.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 125.0mm THK X 2363.5mm X 2975.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Bottom head	[9]	B	22	7		0.727	63.000	1.00	Yes
Shell	[8]	!	-12	-28		0.720	125.000	1.00	Yes
Top head	[9]	B	22	7		0.727	63.000	1.00	Yes
N2400	[1]	B	22	7		0.726	63.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
U1600	[1]	B	22	6		0.726	62.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
K480	[1]	B	22	6		0.726	62.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
K150	[1]	!	-12	-104		0.099	16.764	1.00	No
Nozzle Flg	[5]	A	-18	-38					
K650	[1]	!	-12	-104		0.099	16.764	1.00	No
Nozzle Flg	[5]	A	-18	-38					
K350	[1]	!	-12	-104		0.099	16.764	1.00	No
Nozzle Flg	[5]	A	-18	-38					
M1600	[1]	B	22	7		0.726	63.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
K250	[1]	B	22	7		0.726	63.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
K580	[1]	B	22	7		0.726	63.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
N1400	[1]	B	22	7		0.726	63.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-38					
Nozl. Bolting	[22]		-48			1.000	64.000	1.00	No
Warmest MDMT:			22	7					

Required Minimum Design Metal Temperature -12.0 °C
 Warmest Computed Minimum Design Metal Temperature 7.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and



UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 20 40 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	6.4164	MPa
Design Internal Temperature	400.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-12.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	Y	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16	IP+VE+EW	
Load Case 17	NP+VO+OW	
Load Case 18	FS+BS+IP+OW	
Load Case 19	FS+BS+EP+OW	
Load Case 20	BL+IP+OW	
Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	[V] 97.2	km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	0	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment	H or Hh	0 mm
Distance Upwind of Crest	Lh	0 mm
Distance from Crest to the Vessel	x	0 mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration	[Ss] 0.147
Max. Eff. Ground Acceleration	[S] 0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10



Element To Node	20	
Element Type	Skirt Support	
Description	Skirt-2	
Distance "FROM" to "TO"	2800	mm
Skirt Outside Diameter	3821	mm
Diameter of Skirt at Base	3821	mm
Skirt Thickness	32	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	32	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	50	°C
Design Temperature External Pressure	50	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	MPa
Allowable Stress, Operating	138	MPa
Allowable Stress, Hydrotest	234	MPa
Material Density	0.00775	kg/cm³
P Number Thickness	32.004	mm
Yield Stress, Operating	255.6	MPa
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.5	
Efficiency, Head-to-Skirt or Circ. Seam	0.5	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Weight	
Detail ID	Tailing lug	
Dist. from "FROM" Node / Offset dist	-0.0002896	mm
Miscellaneous Weight	7844.8	N
Offset from Element Centerline	0	mm

Element#	2/5	
Element From Node	20	
Element To Node	30	
Element Type	Skirt Support	
Description	Skirt-1	
Distance "FROM" to "TO"	1000	mm
Skirt Outside Diameter	3821	mm
Diameter of Skirt at Base	3821	mm
Skirt Thickness	32	mm
Internal Corrosion Allowance	0.8	mm
Nominal Thickness	32	mm
External Corrosion Allowance	0.8	mm
Design Temperature Internal Pressure	400	°C
Design Temperature External Pressure	400	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.5	
Weld is pre-Heated	No	

Element#	3/5	
Element From Node	30	
Element To Node	40	
Element Type	Spherical	
Description	Bottom head	
Distance "FROM" to "TO"	0	mm
Inside Diameter	3722	mm
Element Thickness	63	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	70	mm
External Corrosion Allowance	0	mm



Design Internal Pressure	6.4164	MPa
Design Temperature Internal Pressure	400	°C
Design External Pressure	0.1034	MPa
Design Temperature External Pressure	400	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Insulation	
Detail ID	Ins: 10	
Dist. from "FROM" Node / Offset dist	-1825	mm
Height/Length of Insulation	1825	mm
Thickness of Insulation	150	mm
Density	0.000352	kg/cm³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N2400	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	400	mm
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	U1600	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	600	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70 +IT	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	K480	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	80	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B +IT	

Element From Node	30	
Detail Type	Weight	
Detail ID	Closure plate	
Dist. from "FROM" Node / Offset dist	-200	mm
Miscellaneous Weight	1961.2	N
Offset from Element Centerline	0	mm

Element From Node	30	
Detail Type	Weight	
Detail ID	Inert Balls	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	20102	N
Offset from Element Centerline	0	mm



Element#	4/5
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	2975 mm
Inside Diameter	3660 mm
Element Thickness	125 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	125 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	6.4164 MPa
Design Temperature Internal Pressure	400 °C
Design External Pressure	0.1034 MPa
Design Temperature External Pressure	400 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	40
Detail Type	Platform
Detail ID	PL-1
Dist. from "FROM" Node / Offset dist	0 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	360.0
Platform Wind Area	4747520 mm ²
Platform Weight	47020 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	185 mm
Platform Force Coefficient	1.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4300 mm
Unit Weight of Ladder	0.37 N/mm
Platform Length (top head platform)	0 mm
Element From Node	40
Detail Type	Packing
Detail ID	Catalyst
Dist. from "FROM" Node / Offset dist	0 mm
Height of Packed Section	2975 mm
Density	0.000977 kg/cm ³
Percent Volume Holdup	0.0
Specific Gravity of Packing Liquid	0.0
Element From Node	40
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2975 mm
Thickness of Insulation	150 mm
Density	0.000352 kg/cm ³
Element From Node	40
Detail Type	Nozzle
Detail ID	K150
Dist. from "FROM" Node / Offset dist	100 mm
Nozzle Diameter	50.8 mm
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	Y



Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K650	
Dist. from "FROM" Node / Offset dist	2000	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	40	
Detail Type	Nozzle	
Detail ID	K350	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	50.8	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105	+IT
Element From Node	40	
Detail Type	Weight	
Detail ID	Hold grid	
Dist. from "FROM" Node / Offset dist	-0.02558	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	SF wt2	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	21573	N
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Clip wt	
Dist. from "FROM" Node / Offset dist	450	mm
Miscellaneous Weight	16425	N
Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Trunnion 1	
Dist. from "FROM" Node / Offset dist	2970	mm
Miscellaneous Weight	8335.1	N
Offset from Element Centerline	2500	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Trunnion-2	
Dist. from "FROM" Node / Offset dist	2970	mm
Miscellaneous Weight	8335.1	N
Offset from Element Centerline	2500	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	SF wt-1	
Dist. from "FROM" Node / Offset dist	2795	mm
Miscellaneous Weight	21573	N



Offset from Element Centerline	0	mm
Element From Node	40	
Detail Type	Weight	
Detail ID	Hold grid2	
Dist. from "FROM" Node / Offset dist	2970	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	0	mm

Element#	5/5
Element From Node	50
Element To Node	60
Element Type	Spherical
Description	Top head
Distance "FROM" to "TO"	0
Inside Diameter	3722
Element Thickness	63
Internal Corrosion Allowance	3
Nominal Thickness	70
External Corrosion Allowance	0
Design Internal Pressure	6.4164
Design Temperature Internal Pressure	400
Design External Pressure	0.1034
Design Temperature External Pressure	400
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Platform
Detail ID	PL-3
Dist. from "FROM" Node / Offset dist	2200
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	6113280
Platform Weight	28610
Platform Railing Weight	0
Platform Grating Weight	0.017
Platform Width	4000
Platform Height	1000
Platform Clearance or End Offset	2000
Platform Force Coefficient	1.0
Ladder Layout Angle	90.0
Ladder Start Elevation	0
Ladder End Elevation	5200
Unit Weight of Ladder	0.37
Platform Length (top head platform)	4000

Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0
Height/Length of Insulation	1861
Thickness of Insulation	150
Density	0.000352

Element From Node	50
Detail Type	Nozzle
Detail ID	M1600
Dist. from "FROM" Node / Offset dist	0
Nozzle Diameter	600
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	0



Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70	+IT
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K250	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	50	mm
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	+IT
Element From Node	50	
Detail Type	Nozzle	
Detail ID	K580	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	80	mm
Nozzle Schedule	160	
Nozzle Class	600	
Layout Angle	180.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	+IT
Element From Node	50	
Detail Type	Nozzle	
Detail ID	N1400	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	400	mm
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	50	
Detail Type	Weight	
Detail ID	N1Vane distrib	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	2941.8	N
Offset from Element Centerline	0	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	M1Davit	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	0	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Hold down plate	
Dist. from "FROM" Node / Offset dist	-0.02558	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	300	mm
Element From Node	50	
Detail Type	Weight	
Detail ID	Wire meshscreen	
Dist. from "FROM" Node / Offset dist	-0.02558	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	0	mm

Element From Node	50	
Detail Type	Weight	
Detail ID	Inert Balls	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	58836	N
Offset from Element Centerline	0	mm
Element From Node	50	
Detail Type	For./Mom.	
Detail ID	F&M	
Dist. from "FROM" Node / Offset dist	1825	mm
Force in X Direction	21440	N
Force in Y Direction	-29686	N
Force in Z Direction	21440	N
Moment about X Axis	19806728	N-mm
Moment about Y Axis	0	N-mm
Moment about Z Axis	19806728	N-mm
Force/Moment Combination Method	SRSS	



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
	Skirt-2	...	2800	...	2800
	Skirt-1	...	3800	...	1000
	Bottom head	...	3800	...	...
	Shell	...	6775	...	2975
	Top head	...	6775	...	...

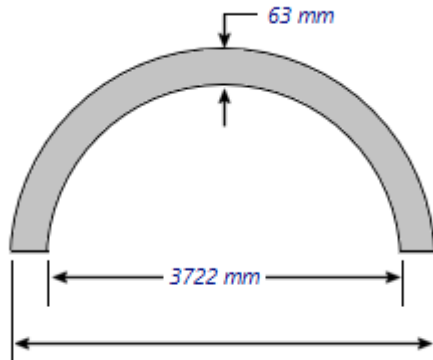
The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Bottom head

Spherical Head Design Information:					
Design Pressure	P	6.416 MPa	Int. Design Temperature		400 °C
Material		SA-516 70	Ext. Design Temperature		400 °C
Length		0.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	101.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3722.000 mm	Actual Stress		100.3 MPa
Spec. Min. Thk t _e , or s _a		63.0000 mm	Spec. Nominal Thk.		70.000 mm
Surface Area		23428496.0 mm ²	SG of Contents		0.000
Element Volume		13501.3 ltr	Weight of Contents		0.0 N
Empty Weight		120178.1 N	Operating Weight		120178.1 N
			Comp. All. Ope. Stress		82.5 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 46.525 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 23.263 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (6.416 \cdot 1864.0) / (2 \cdot 101.03 \cdot 1.0 - 0.2 \cdot 6.416)$$

$$= 59.5707 + 3.0000 = 62.5707 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 101.03 \cdot 1.0 \cdot 60.0) / (1864.0 + 0.2 \cdot 60.0)$$

$$= 6 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 63.0) / (1861.0 + 0.2 \cdot 63.0)$$

$$= 9 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (6.416 \cdot (1864.0 + 0.2 \cdot 60.0)) / (2 \cdot 1.0 \cdot 60.0)$$

= 100.309 MPa

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 2.769 %

MDMT Calculations in the Spherical Portion:

Govrn. thk, $t_g = 63.0$, $t_r = 43.597$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.727$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 22 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 7 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.425 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	63.000 mm	Basic MDMT	22 °C
Stress/Thk. Ratio	0.727	Temperature Difference	15 °C
UCS-66(c) red.	No	Computed Min. Temp.	7 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 46.525 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 23.263 mm

EXTERNAL PRESSURE RESULTS for: Bottom head

Spherical Head From 30 to 40 Ext. Chart: CS-2 at 400 °C

Elastic Modulus from Chart: CS-2 at 400 °C: 162839 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Sphere OD	Do/t	Factor A	Factor B
60.000	3848.00	64.13	0.0038981	68.77

$$MAEP = B / ((Do/t)/2) = 68.7744 / (64.1333/2) = 2.1447 \text{ MPa}$$

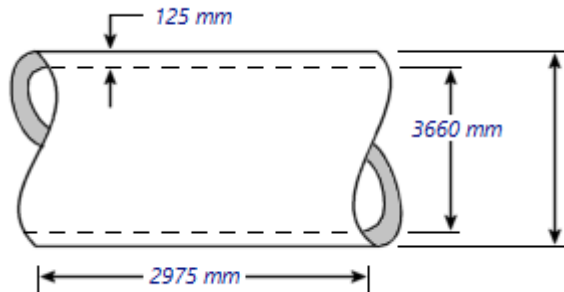
Results for Required Thickness				Tca
Tca	Sphere OD	Do/t	Factor A	Factor B
6.134	3848.00	627.37	0.0003985	32.44

$$MAEP = B / ((Do/t)/2) = 32.4447 / (627.3732/2) = 0.1034 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell

Cylindrical Shell Design Information:					
Design Pressure	P	6.416 MPa	Int. Design Temperature		400 °C
Material		SA-516 70	Ext. Design Temperature		400 °C
Length		2975.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	101.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3660.000 mm	Actual Stress		100.3 MPa
Spec. Min. Thk t _e , or s _a		125.0000 mm	Spec. Nominal Thk.		125.000 mm
Surface Area		36543792.0 mm ²	SG of Contents		0.000
Element Volume		31305.2 ltr	Weight of Contents		0.0 N
Empty Weight		336073.4 N	Operating Weight		336073.4 N
			Comp. All. Ope. Stress		92.2 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +IT

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (6.416 \cdot 1833.0) / (101.03 \cdot 1.0 - 0.6 \cdot 6.416)$$

$$= 121.0273 + 3.0000 = 124.0273 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (101.03 \cdot 1.0 \cdot 122.0) / (1833.0 + 0.6 \cdot 122.0)$$

$$= 6 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 125.0) / (1830.0 + 0.6 \cdot 125.0)$$

$$= 9 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (6.416 \cdot (1833.0 + 0.6 \cdot 122.0)) / (1.0 \cdot 122.0)$$

$$= 100.253 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 3.303 %

Minimum Design Metal Temperature Results:



Note: This Element/Detail was specified as being Impact Tested.

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +IT

EXTERNAL PRESSURE RESULTS for: Shell

Cylindrical Shell From 40 to 50 Ext. Chart: CS-2 at 400 °C

Elastic Modulus from Chart: CS-2 at 400 °C: 162839 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
122.000	3910.00	4215.67	32.05	1.0782	0.0070508	75.65

$$MAEP = (4*B)/(3*(Do/t)) = (4*75.6534)/(3*32.0492) = 3.1474 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.045	3910.00	4215.67	278.39	1.0782	0.0002652	21.60

$$MAEP = (4*B)/(3*(Do/t)) = (4*21.5957)/(3*278.3923) = 0.1034 \text{ MPa}$$

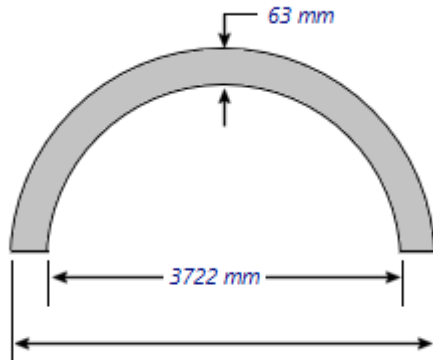
Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
122.000	3910.00	109008.18	32.05	27.8793	0.0010922	53.10

$$MAEP = (4*B)/(3*(Do/t)) = (4*53.0966)/(3*32.0492) = 2.209 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Top head

Spherical Head Design Information:					
Design Pressure	P	6.416 MPa	Int. Design Temperature		400 °C
Material		SA-516 70	Ext. Design Temperature		400 °C
Length		0.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	101.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	3722.000 mm	Actual Stress		100.3 MPa
Spec. Min. Thk t _e , or s _a		63.0000 mm	Spec. Nominal Thk.		70.000 mm
Surface Area		23428496.0 mm ²	SG of Contents		0.000
Element Volume		13501.3 ltr	Weight of Contents		0.0 N
Empty Weight		120178.1 N	Operating Weight		120178.1 N
			Comp. All. Ope. Stress		82.5 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 46.525 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 23.263 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ per UG-27 (d)}$$

$$= (6.416 \cdot 1864.0) / (2 \cdot 101.03 \cdot 1.0 - 0.2 \cdot 6.416)$$

$$= 59.5707 + 3.0000 = 62.5707 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 101.03 \cdot 1.0 \cdot 60.0) / (1864.0 + 0.2 \cdot 60.0)$$

$$= 6 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (R + 0.2 \cdot t) \text{ per UG-27 (d)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 63.0) / (1861.0 + 0.2 \cdot 63.0)$$

$$= 9 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (6.416 \cdot (1864.0 + 0.2 \cdot 60.0)) / (2 \cdot 1.0 \cdot 60.0)$$

= 100.309 MPa

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom}/R_f) \cdot (1 - R_f/R_o)$ 2.769 %

MDMT Calculations in the Spherical Portion:

Govrn. thk, $t_g = 63.0$, $t_r = 43.597$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^*/(t_g - c) = 0.727$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 22 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 7 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	6.425 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	63.000 mm	Basic MDMT	22 °C
Stress/Thk. Ratio	0.727	Temperature Difference	15 °C
UCS-66(c) red.	No	Computed Min. Temp.	7 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 46.525 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 23.263 mm

EXTERNAL PRESSURE RESULTS for: Top head

Spherical Head From 50 to 60 Ext. Chart: CS-2 at 400 °C

Elastic Modulus from Chart: CS-2 at 400 °C: 162839 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Sphere OD	Do/t	Factor A	Factor B
60.000	3848.00	64.13	0.0038981	68.77

$$MAEP = B/((Do/t)/2) = 68.7744/(64.1333/2) = 2.1447 \text{ MPa}$$

Results for Required Thickness				Tca
Tca	Sphere OD	Do/t	Factor A	Factor B
6.134	3848.00	627.37	0.0003985	32.44

$$MAEP = B/((Do/t)/2) = 32.4447/(627.3732/2) = 0.1034 \text{ MPa}$$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	11.084 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	11.069 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	11.735 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	9.379 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	10.643 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	11.069 MPa
App 27-4	= M.A.W.P.	6.425 MPa

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	184.0	234.0	0.786	11.77
Shell	183.9	234.0	0.786	11.77
Top head	184.0	234.0	0.786	11.77

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2_400	Nozzle	118.00	88.90	1.327
N2_400	Pad	138.00	101.00	1.366
U1_600	Nozzle	138.00	101.00	1.366
U1_600	Pad	138.00	101.00	1.366
K4_80	Nozzle	118.00	88.92	1.327
K4_80	Pad	138.00	101.03	1.366
K1_50	Nozzle	138.00	101.02	1.366
K6_50	Nozzle	138.00	101.02	1.366
K3_50	Nozzle	138.00	101.02	1.366
M1_600	Nozzle	138.00	101.00	1.366
M1_600	Pad	138.00	101.00	1.366
K2_50	Nozzle	118.00	88.90	1.327
K2_50	Pad	138.00	101.00	1.366
K5_80	Nozzle	118.00	88.90	1.327
K5_80	Pad	138.00	101.00	1.366
N1_400	Nozzle	118.00	88.90	1.327
N1_400	Pad	138.00	101.00	1.366

Minimum 1.327

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	101.03	1.366
Shell	138.00	101.03	1.366
Top head	138.00	101.03	1.366

Minimum 1.366

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N2_400	125.20	216.00	0.580
U1_600	181.16	179.40	1.010
K4_80	53.08	216.00	0.246
K1_50	31.35	225.00	0.139
K6_50	31.35	225.00	0.139
K3_50	31.35	225.00	0.139



M1_600	181.16	179.40	1.010
K2_50	30.95	216.00	0.143
K5_80	72.99	216.00	0.338
N1_400	125.20	216.00	0.580

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	11.084 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	11.069 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	11.724 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	9.379 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	10.643 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	11.069 MPa
App 27-4	= M.A.W.P.	6.425 MPa

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 6.416 • 1.327
 = 11.069 MPa

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Bottom head	174.1	234.0	0.744	11.13
Shell	173.7	234.0	0.742	11.12
Top head	173.3	234.0	0.741	11.09

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2_400	Nozzle	118.00	88.90	1.327
N2_400	Pad	138.00	101.00	1.366
U1_600	Nozzle	138.00	101.00	1.366
U1_600	Pad	138.00	101.00	1.366
K4_80	Nozzle	118.00	88.92	1.327
K4_80	Pad	138.00	101.03	1.366
K1_50	Nozzle	138.00	101.02	1.366
K6_50	Nozzle	138.00	101.02	1.366
K3_50	Nozzle	138.00	101.02	1.366
M1_600	Nozzle	138.00	101.00	1.366
M1_600	Pad	138.00	101.00	1.366
K2_50	Nozzle	118.00	88.90	1.327
K2_50	Pad	138.00	101.00	1.366
K5_80	Nozzle	118.00	88.90	1.327
K5_80	Pad	138.00	101.00	1.366
N1_400	Nozzle	118.00	88.90	1.327
N1_400	Pad	138.00	101.00	1.366

Minimum 1.327

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Bottom head	138.00	101.03	1.366
Shell	138.00	101.03	1.366
Top head	138.00	101.03	1.366

Minimum 1.366

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N2_400	118.43	216.00	0.548



U1_600	171.36	179.40	0.955
K4_80	50.21	216.00	0.232
K1_50	29.61	225.00	0.132
K6_50	29.61	225.00	0.132
K3_50	29.61	225.00	0.132
M1_600	170.63	179.40	0.951
K2_50	29.15	216.00	0.135
K5_80	68.74	216.00	0.318
N1_400	117.93	216.00	0.546

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	3800	...	...	No Calc	No Calc
20	30	3800	...	...	No Calc	No Calc
30	40	No Calc	3848	60	0.0038981	68.7744
40	50	4215.67	3910	122	0.0070508	75.6534
50	60	No Calc	3848	60	0.0038981	68.7744

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	63	9.13351	0.10343	2.14473
40	50	125	17.0449	0.10343	3.14739
50	60	63	9.13351	0.10343	2.14473

Minimum

2

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	3800	No Calc	No Calc	No Calc
20	30	3800	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	4215.67	109008	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	9399.89	...	8986.59	...	939.989
20	30	2952.26	...	2804.65	...	295.226
30	40	12255.6	13501.3	11748.8	13566.7	1225.56
40	50	34272.2	31305.2	33476.2	31408	3427.22
50	60	12255.6	13501.3	11748.8	13566.7	1225.56
Total		71135	58307.83	68765	58541.35	7113

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Wght	800	...	-0.00028956	...	Tailing lug
30	Insl	1401.46	...	-912.5	...	Ins: 10
30	Nozl	681.717	...	-1779.5	...	N2_400
30	Nozl	2294.4	-1500	-1320.01	...	U1_600
30	Nozl	48.3069	1500	-1165.26	...	K4_80
30	Wght	200	...	-200	...	Closure plate
30	Wght	2050	...	...	...	Inert Balls
40	Plat	5274.52	0.25084E-04	...	...	PL-1
40	Pack	30615.5	...	1487.5	...	Catalyst
40	Insl	2005.88	...	1487.5	...	Ins: 20
40	Nozl	29.1004	1980	100	...	K1_50
40	Nozl	29.1004	1980	2000	...	K6_50
40	Nozl	23.0331	-1980	2500	...	K3_50
40	Wght	100	...	-0.025579	...	Hold grid
40	Wght	2200	...	...	...	SF wt2
40	Wght	1675	...	450.004	...	Clip wt
40	Wght	850	2500	2969.97	...	Trunnion 1
40	Wght	850	-2500	2969.97	...	Trunnion-2
40	Wght	2200	...	2794.97	...	SF wt-1
40	Wght	100	...	2969.97	...	Hold grid2
50	Plat	3209.37	...	2200	...	PL-3
50	Insl	1401.46	...	930.5	...	Ins: 50
50	Nozl	1687.75	...	2054.51	...	M1_600
50	Nozl	49.4479	1500	1470.01	...	K2_50
50	Nozl	83.099	-1500	1670.01	...	K5_80
50	Nozl	681.717	1500	1020.01	...	N1_400
50	Wght	300	...	...	...	N1_Vane distrib
50	Wght	100	...	...	...	M1_Davit
50	Wght	100	300	-0.025579	...	Hold down plate
50	Wght	100	...	-0.025579	...	Wire meshscreen
50	Wght	6000	...	...	...	Inert Balls
50	Forc	...	...	1825	...	F&M

Total Weight of Each Detail Type:

Platforms	8483.9
Packing	30615.5
Insulation	4808.8
Nozzles	5607.7
Weights	17625.0

Sum of the Detail Weights 67140.9 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	78249.1	78249.1	78249.1	78249.1	78249.1	78249.1



Nozzles	5607.7	5607.7	5607.7	5607.7	5607.7	5607.7
Wld Weights	8875.0	8875.0	8875.0	8875.0	8875.0	8875.0
Platforms	...	...	8483.9	8483.9	8483.9	8483.9
Packing	...	...	30615.5	30615.5	30615.5	30615.5
Insulation	...	...	4808.8	4808.8	4808.8	4808.8
Empty Weights	...	...	...	700.0	700.0	...
Test Weights	...	200.0	...	...	...	...
Ope Weights	...	...	...	...	...	8750.0
Test Liquid	...	58272.2	...	...	...	...
Totals	92731.8	151204.0	136640.0	137340.0	137340.0	145390.0

Field Installation Options:**Miscellaneous Weight Percent: 10.0 %**

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	92731.8 kgm
Shop Test Wt.	- Fabricated weight + water (full)	151204.0 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	136640.0 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	137340.0 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	137340.0 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	145390.0 kgm
Field Test Wt.	- Empty weight + test medium (full)	162122.5 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	85747.5 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	89171.1 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	80969.6 kgm

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	33611272	33048300
20	30	12004025	11802964
30	40	23428496	21760686
40	50	36543792	34207236
50	60	23428496	21760686

Total	129016088.0	122579872.0 mm ²
Total	129.0	122.6 m ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	11139.9	11139.9	10685.2	...	11139.9
20	30	3247.49	3247.49	3085.11	...	3247.49
30	40	18107	20157	31108	33051128	20157
40	50	83651.6	83651.6	83549.3	683089	83651.6
50	60	21194	27194	33694.9	9830556	27194

Empty Support Force + the Sum of the Y forces	1376441.62	N
Operating Support Force + the Sum of the Y forces	1455380.00	N
Hydro Support Force + the Sum of the Y forces	1619459.00	N

**Cumulative Vessel Weight**

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	145390	145390	162122
20	30	134250	134250	151437
30	40	131003	131003	148352
40	50	110846	110846	117244
50	60	27194	27194	33694.9

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	43564772	43564772	43564772
20	30	43564772	43564772	43564772
30	40	43564772	43564772	43564772
40	50	10513645	10513645	10513645
50	60	9830556	9830556	9830556

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Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N2_400	6.93	10.20	400	600	GR 1.1	...	...	...	...
U1_600	6.93	10.20	400	600	GR 1.1	...	...	...	...
K4_80	6.93	10.20	400	600	GR 1.1	...	...	...	...
K1_50	6.93	10.20	400	600	GR 1.1	...	...	...	...
K6_50	6.93	10.20	400	600	GR 1.1	...	...	...	...
K3_50	6.93	10.20	400	600	GR 1.1	...	...	...	...
M1_600	6.93	10.20	400	600	GR 1.1	...	...	...	...
K2_50	6.93	10.20	400	600	GR 1.1	...	...	...	...
K5_80	6.93	10.20	400	600	GR 1.1	...	...	...	...
N1_400	6.93	10.20	400	600	GR 1.1	...	...	...	...
Min Rating	6.931	10.201, [for Core Elements]					...	...	...

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 15.9403 Hz.

The Natural Frequency for the Vessel (Ope...) is 15.9403 Hz.

The Natural Frequency for the Vessel (Filled) is 15.4057 Hz.

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Forces/Moments Applied to Vessel (Combined w/Wind Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	30320.8	83368800

Forces/Moments Applied to Vessel (Combined w/Seismic Loads)

From	To	X and Z Dir Force Res. N	X,Z Moment and For Res N-mm
10	20	...	...
20	30	...	...
30	40	...	...
40	50	...	...
50	60	30320.8	83368800

User Input Forces and Moments:

From Node	Distance From	- - - Fx	Forces - - - Fy	- - - Fz	- - - Mx	Moments - - - My	- - - Mz
50	1825.00	21440.	-29686.	21440.	0.198E+08	0.198E+08	

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.228 \cdot 0.941) / (1 + 1.7 \cdot 3.4 \cdot 0.228)))$
 $= \min(0.85, 0.894)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	15.940 Hz
Natural Frequency of Vessel (Empty)	15.940 Hz
Natural Frequency of Vessel (Test)	15.406 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.522
Structure Height to Diameter ratio	2.317
Height to top of Structure	8849.000 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$; $Z_g = 274320.0$ mm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1400.0 + 0.0 = 1400.0$ mm
 $= 4.593$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[K_z]$:
 Because z (4.593 ft.) < 15 ft.
 $= 2.01(15 / Z_g)^{2/\alpha}$
 $= 2.01(15 / 274320.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[K_{zt}]$:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf [0.766] kPa}$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.522 \cdot 138.193$$

$$= 981.0 \text{ lbs. [4363.3] N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Skirt-2	1400.0	0.000	0.000	0.000	0.849	1.000	0.766
Skirt-1	3300.0	0.000	0.000	0.000	0.849	1.000	0.766
Bottom head	3800.0	0.000	0.000	0.000	0.849	1.000	0.766
Shell	5287.5	0.000	0.000	0.000	0.875	1.000	0.766
Top head	7655.2	0.000	0.000	0.000	0.946	1.000	0.766

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure mm
- Lc - Total length of conical section(s) of vessel, mm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, N
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
- Z - Maximum amplitude of vibration at top of vessel, mm
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.000003 (High); 0.000003 < 0.000004 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0.0/6775.0 = 0.0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[13.1/22.23^2] \cdot 10^4 = 265.074$ [Geometry Violation]

Compute the vibration possibility. If Vp > 0.000004 no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 1400124 / (6775.0 \cdot 4074.808^2)$$

$$= 0.12446E-04$$

Since Vp is > 0.000004 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area	Elevation	Pressure	Force	Cf
----	-----------	-----------	----------	-------	----



	mm ²	mm	kPa	N	
PL-1	4747520.00	3800.00	0.77	3636.82	1.00
PL-3	6113280.00	8974.97	0.77	4683.05	1.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Tailing lug	0.00	-0.00	0.77	0.00
Closure plate	0.00	3599.97	0.77	0.00
Inert Balls	0.00	3799.97	0.77	0.00
Hold grid	0.00	3799.97	0.77	0.00
SF wt2	0.00	3800.00	0.77	0.00
Clip wt	0.00	4250.00	0.77	0.00
Trunnion 1	0.00	6769.98	0.77	0.00
Trunnion-2	0.00	6769.98	0.77	0.00
SF wt-1	0.00	6594.97	0.77	0.00
Hold grid2	0.00	6769.98	0.77	0.00
N1Vane distr	0.00	6774.97	0.77	0.00
M1Davit	0.00	6774.97	0.77	0.00
Hold down pla	0.00	6774.94	0.77	0.00
Wire meshscre	0.00	6774.94	0.77	0.00
Inert Balls	0.00	6774.97	0.77	0.00

The Natural Frequency for the Vessel (Ope...) is 15.9403 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1400	4585.2	12838560	0.76606	4799.61
20	30	3300	4585.2	4585200	0.76606	1714.15
30	40	3799.98	4977.6	151.717	0.76606	0.056719
40	50	5287.5	5052	15029701	0.76606	8493.82
50	60	7655.22	4977.6	8108243	0.76606	9070.67

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	2.000
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned}SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104\end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
= $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
= $0.020(28.33330.75)$
= 0.246 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
= (1/Natural Frequency) = (1/15.94)
= 0.063

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
= Minimum Value of ($1.7 \cdot 0.246$, 0.063) per 12.8.2
= 0.063

Seismic Response Coefficient per equation 12.8-2 [Cs]:
= $Sds / (R / I)$
= $0.157 / (2.0 / 2.0)$
= 0.157

Maximum value of C_s per equation 12.8-3 [Cs]:
= $Sd1 / (T(R / I_e))$
= $0.104 / (0.063(2.0 / 2.0))$
= 1.658

Minimum value of C_s per equation 15.4-1 [Cs]:
= $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
= $\max((0.044 \cdot 0.1568 \cdot 2.0), 0.030)$
= 0.030

Total Base Shear [V]:
= $C_s \cdot W$ (Equation 12.8-1):
= $0.1568 \cdot 1425694$
= 223548.797 N

Vertical load per 12.4-4, [Ev]:
= $0.2 \cdot Sds \cdot W$
= $0.2 \cdot 0.157 \cdot 1425694$

= 44709.8 N

Distribute the Base shear force to each element according to the equations
 $F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor
 $C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related
 to the period of Vibration.

In this case, the value of k was 1.0

The Natural Frequency for the Vessel (Ope...) is 15.9403 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	1400	109238	4779.49	4836	15023.5
20	30	3300	31844.9	3284.24	3323.07	11597.8
30	40	3799.98	197660	23473.7	21335.7	10599.1
40	50	5287.5	820288	135550	137152	4400.52
50	60	6774.98	266664	56461.8	44524.4	-21323.7



Bending Moments due to user defined forces and moments.

User Force/Moment Shear and Bending

From	To	Distance to Support mm	Cumulative Shr Wind Cas N	Cumulative Shr Eqk Cas N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1400	30320.8	30320.8	288874432	288874432
20	30	3300	30320.8	30320.8	203941840	203941840
30	40	3799.98	30320.8	30320.8	173609696	173609696
40	50	5287.5	30320.8	30320.8	173608752	173608752
50	60	6774.98	30320.8	30320.8	83368792	83368792

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1400	24078.3	223549	126776744	1256183680
20	30	3300	19278.7	218769	66052336	636687360
30	40	3799.98	17564.5	215485	47623784	419478720
40	50	5287.5	17564.5	192011	47623244	419472416
50	60	6774.98	9070.67	56461.8	7987661	49720448

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Applied Forces Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	30320.8	0.0020014	0.007593	0.51289E-05
20	30	30320.8	0.010343	0.013447	0.65419E-05
30	40	30320.8	0.013447	0.013447	0.65419E-05
40	50	30320.8	0.023514	0.034125	0.72735E-05
50	60	30320.8	0.034125	0.034125	0.72735E-05

Allowable deflection at the Tower Top (For)(6.000"/100ft. Criteria)
 Allowable deflection : 33.875 Actual deflection : 0.034 mm

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	24078.3	0.00084667	0.0030826	0.19904E-05
20	30	19278.7	0.0041472	0.005337	0.24912E-05
30	40	17564.5	0.005337	0.0053371	0.24912E-05
40	50	17564.5	0.0091413	0.013071	0.26639E-05
50	60	9070.67	0.013072	0.013072	0.26639E-05

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	1312.06	8200.4
20	30	1312.06	8200.4
30	40	1424.35	8902.19
40	50	1445.64	9035.25
50	60	1424.35	8902.19

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 33.875 Actual deflection : 0.013 mm

Total Deflection in the Operating Condition + Applied Forces:
 Allowable deflection : 33.875 Actual deflection : 0.047 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	380911	361866	357822336	340072320
20	30	380911	361866	357822336	340072320
30	40	749128	714021	697450752	665801920
40	50	1486365	1451841	1362994048	1333376768
50	60	749128	714021	697450752	665801920

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Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	82.7985	140.397	-124.37	-124.37
20	30	60.6165	140.397	-72.7716	-124.37
30	40	121.233	280.795	-82.5293	-139.357
40	50	121.234	280.795	-92.1593	-147.277
50	60	121.233	280.795	-82.5293	-139.357

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	98.3844	-1.68451	169.725
40	50	46.9184	-0.85536	80.9398
50	60	98.3844	-1.68451	169.725

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-3.94017	-3.94017	-4.39363	0.12806	0.12806
20	30	-3.63827	-3.63827	-4.10406	0.12806	0.12806
30	40	-1.79928	-1.79928	-1.79928	0.065411	0.065411
40	50	-0.74873	-0.74873	-0.74873	0.0078825	0.0078825
50	60	-0.3735	-0.3735	-0.3735	0.01476	0.01476

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	0.12806	0.37267	3.69269	0.18634	...
20	30	0.12806	0.19417	1.87161	0.097084	...
30	40	0.065411	0.071506	0.62983	0.035753	...
40	50	0.0078825	0.035705	0.31449	0.017852	...
50	60	0.01476	0.011993	0.074654	0.0059966	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	3.42559
20	30	...	...	...	1.70666
30	40	...	...	...	0.56429
40	50	...	...	...	0.28176
50	60	...	...	...	0.05887

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	-0.082044	0.04152
20	30	-0.082044	0.032053
30	40	-0.04158	0.014846

40	50	-0.020449	0.0030312
50	60	-0.04158	-0.029867

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	0.84918	0.84918	0.80705	0.80705
20	30	0.59951	0.59951	0.56977	0.56977
30	40	0.26067	0.26067	0.24884	0.24884
40	50	0.13016	0.13016	0.12733	0.12733
50	60	0.12518	0.12518	0.1195	0.1195

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...

Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		82.80	-5.37	124.37		0.0432
20		60.62	-4.64	72.77		0.0638
30		121.23	-2.24	82.53		0.0271
40		121.23	-0.94	92.16		0.0102
50		121.23	-0.57	82.53		0.0069

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.50	82.80	-8.30	124.37	0.0061	0.0667
20		60.62	-6.04	72.77		0.0830
30		121.23	-2.67	82.53		0.0324
40		121.23	-1.17	92.16		0.0126
50		121.23	-0.60	82.53		0.0073

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		82.80	-5.37	124.37		0.0432
20		60.62	-4.64	72.77		0.0638
30		121.23	-2.24	82.53		0.0271
40		121.23	-0.94	92.16		0.0102



50 121.23 -0.57 82.53 0.0069

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.77	82.80	-8.57	124.37	0.0093	0.0689
20		60.62	-6.21	72.77		0.0853
30		121.23	-2.74	82.53		0.0332
40		121.23	-1.20	92.16		0.0130
50		121.23	-0.62	82.53		0.0075

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.40	-4.71	124.37		0.0379
20		140.40	-4.33	124.37		0.0348
30		280.79	-1.90	139.36		0.0136
40		280.80	-0.77	147.28		0.0053
50		280.79	-0.39	139.36		0.0028

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.40	-4.52	124.37		0.0364
20		140.40	-4.23	124.37		0.0340
30		280.79	-1.86	139.36		0.0134
40		280.80	-0.76	147.28		0.0051
50		280.79	-0.39	139.36		0.0028

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		82.80	-5.37	124.37		0.0432
20		60.62	-4.64	72.77		0.0638
30	96.94	121.23		82.53	0.7996	
40	46.32	121.23		92.16	0.3821	
50	98.12	121.23		82.53	0.8094	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.77	82.80	-8.57	124.37	0.0093	0.0689
20		60.62	-6.21	72.77		0.0853
30	97.56	121.23		82.53	0.8047	
40	46.63	121.23		92.16	0.3846	
50	98.20	121.23		82.53	0.8100	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		82.80	-5.37	124.37		0.0432
20		60.62	-4.64	72.77		0.0638
30		121.23	-3.92	82.53		0.0475
40		121.23	-1.80	92.16		0.0195
50		121.23	-2.25	82.53		0.0273

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	0.77	82.80	-8.57	124.37	0.0093	0.0689
20		60.62	-6.21	72.77		0.0853
30		121.23	-4.42	82.53		0.0536
40		121.23	-2.05	92.16		0.0223



50 121.23 -2.30 82.53 0.0279

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.40	-4.71	124.37		0.0379
20		140.40	-4.33	124.37		0.0348
30	168.03	280.79		139.36	0.5984	
40	80.22	280.80		147.28	0.2857	
50	169.37	280.79		139.36	0.6032	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		140.40	-4.52	124.37		0.0364
20		140.40	-4.23	124.37		0.0340
30	167.99	280.79		139.36	0.5983	
40	80.20	280.80		147.28	0.2856	
50	169.37	280.79		139.36	0.6032	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		82.80	-4.07	124.37		0.0327
20		60.62	-3.77	72.77		0.0518
30	96.65	121.23		82.53	0.7972	
40	46.18	121.23		92.16	0.3809	
50	98.03	121.23		82.53	0.8086	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		82.80	-3.74	124.37		0.0301
20		60.62	-3.46	72.77		0.0475
30	96.67	121.23		82.53	0.7974	
40	46.19	121.23		92.16	0.3810	
50	98.03	121.23		82.53	0.8086	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		82.80	-4.07	124.37		0.0327
20		60.62	-3.77	72.77		0.0518
30	96.65	121.23		82.53	0.7972	
40	46.18	121.23		92.16	0.3809	
50	98.03	121.23		82.53	0.8086	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		82.80	-4.07	124.37		0.0327
20		60.62	-3.77	72.77		0.0518
30	96.65	121.23		82.53	0.7972	
40	46.18	121.23		92.16	0.3809	
50	98.03	121.23		82.53	0.8086	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		82.80	-4.07	124.37		0.0327
20		60.62	-3.77	72.77		0.0518
30		121.23	-1.86	82.53		0.0226
40		121.23	-0.76	92.16		0.0082

50 121.23 -0.39 82.53 0.0047

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		69.00	-4.88	103.64		0.0470
20		50.51	-4.33	60.64		0.0715
30	96.93	101.03		68.77	0.9594	
40	46.31	101.03		76.80	0.4584	
50	98.12	101.03		68.77	0.9712	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		69.00	-4.88	103.64		0.0470
20		50.51	-4.33	60.64		0.0715
30		101.03	-3.80	68.77		0.0552
40		101.03	-1.74	76.80		0.0226
50		101.03	-2.23	68.77		0.0324

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		82.80	-4.07	124.37		0.0327
20		60.62	-3.77	72.77		0.0518
30	96.65	121.23		82.53	0.7972	
40	46.18	121.23		92.16	0.3809	
50	98.03	121.23		82.53	0.8086	

Absolute Maximum of the all of the Stress Ratio's 0.9712

Governing Element: Top head

Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-2	...	-3.9	0.1	0.8	...	+ Tens
Skirt-2	...	-3.9	-0.1	-0.8	-4.9	- Comp
Skirt-1	...	-3.6	0.1	0.6	...	+ Tens
Skirt-1	...	-3.6	-0.1	-0.6	-4.3	- Comp
Bottom head	98.4	-1.8	0.1	0.3	96.9	+ Tens
Bottom head	98.4	-1.8	-0.1	-0.3	...	- Comp
Shell	46.9	-0.7	...	0.1	46.3	+ Tens
Shell	46.9	-0.7	...	-0.1	...	- Comp
Top head	98.4	-0.4	...	0.1	98.1	+ Tens
Top head	98.4	-0.4	...	-0.1	...	- Comp

**Skirt Support Analysis Per EN 13445-3 Section 16.12:**

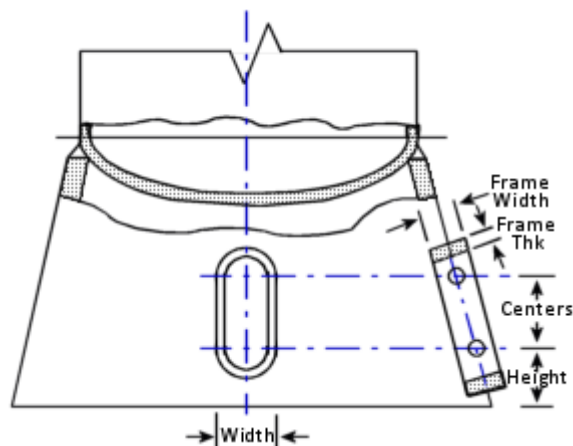
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

The skirt comprises more than one piece. If this is the case, the Openings will only be considered in this Section of the Skirt! Otherwise, make the whole skirt all one piece.

Item: Skirt-2

Total vessel (empty) weight	Fg	1304568.625 N
Weight of vessel contents (liquid)	Ff	0.000 N
Vessel Gross Weight	Fg + Ff	1304568.625 N
Skirt Mean Diameter	Dz	3789.000 mm
Skirt Analysis Thickness (Corroded)	ez	30.400 mm
Skirt Height	H	2800.000 mm
Additional Axial Force at section 1-1	F1	-29686.283 N
Moment at section 1-1	M1	840629184.000 N-mm
Applied Moment - Bottom of Skirt	Mb	1545058176.000 N-mm
Weight at Bottom of Skirt	W	1425693.750 N



Access Opening: 2 Cuts through the top of the skirt.

Note:

If an opening cuts the skirt top or bottom, this is not a problem. This becomes a 'Sentry Box' or 'Mouse Hole' type Opening.

Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Oround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	600.0000	0.0	1500.00	0.00	50.000	16.000
2	1000.0000	180.0	2800.00	0.00	50.000	18.000

Interference check:

Access	Access	Pitch mm	Ligament mm
1	2	6091.9707	5291.9707
1	2	6091.9707	5291.9707

Interference Check: No Overlap Detected between Openings

Minimum Pitch between openings is: 6091.9707 mm
 Minimum Ligament between Openings is: 5291.9707 mm

Recommended minimum Ligament [IMin]:

$$= 2 \cdot \sqrt{(D \cdot t)}$$

$$= 2 \cdot \sqrt{(3789.0 \cdot 30.4)} \text{ mm}$$

$$= 678.780 \text{ mm}$$

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

Number of Openings At Critical Height = 2

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
2	164.6971	195.3029

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
0.000	164.697
-164.697	0.000

Height of Weakest Section	H	2799.902	mm
Rotation of Principle Axes	a	-0.000	deg.
Centroid Position in X direction	g	171.404	mm
Centroid Position in Y direction	h	-0.000	mm
Principle Moment of Inertia	Ixx	647237795840.000	mm ⁴
Principle Moment of Inertia	Iyy	543489097728.000	mm ⁴
Distance to Extreme Fiber	ybar	1909.700	mm
Distance to Extreme Fiber	xbar	2081.104	mm
Minimum Section Modulus	W4	261154512.000	mm ³
Metal Area at the Cutting Plane	A4	334501.781	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / D_z)$$

$$= \sin^{-1}(1000.0/3789.0)$$

$$= 0.2671 \text{ rad (15.3029 deg)}$$

Note: The largest opening width is considered.

Skirt Mean Radius at Section 4-4 [r]:

$$= D_z / 2$$

$$= 3789.0/2$$

$$= 1894.5 \text{ mm}$$

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.

Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
1600.0	17824.7	...	...
1800.0	30197.5	...	...

This calculation of ratio below is per an earlier version of EN 13445.

Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$= \delta \cdot \sqrt{(r / e_z)}$$

$$= 0.267 \cdot \sqrt{(1894.5/30.4)}$$

$$= 2.108$$

WARNING: ratio Exceeds 2 - See EN 13445 16.12.76(a).
 This equation is a precaution against radial buckling at the opening.
 The frame (if there is one), may prevent this. The code does not



say what size this frame should be. Your judgement is required.

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$\begin{aligned} &= ex(\text{new}) - ci - cext \\ &= 32.0 - 0.8 - 0.8 \\ &= 30.4 \text{ mm} \end{aligned}$$

Note: At least one of the openings cuts the Top Of the Skirt (Ok)

Find the moment acting at the Critical Height [M4]:

$$\begin{aligned} &= (M_{\text{bottom}} - M_{\text{top}}) \cdot (H - \text{CutHeight}) / H + M_{\text{top}} \\ &= (.15451\text{E}+10 - 0.84063\text{E}+09) \cdot (2800.0 - 2800.0) / 2800.0 + 0.84063\text{E}+09 \\ &= 0.84063\text{E}+09 \text{ N-mm} \end{aligned}$$

Compute the section modulus [Zxx and Zyy]:

$$\begin{aligned} Z_{xx} &= I_{xx} / y_{\text{bar}} = 0.64724\text{E}+12 / 1909.7 = 0.33892\text{E}+09 \text{ mm}^3 \\ Z_{yy} &= I_{yy} / x_{\text{bar}} = 0.54349\text{E}+12 / 2081.104 = 0.26115\text{E}+09 \text{ mm}^3 \\ Z &= \min(Z_{xx}, Z_{yy}) = \min(.33892\text{E}+09, .26115\text{E}+09) = 0.26115\text{E}+09 \text{ mm}^3 \end{aligned}$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned} &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area}(A_4))} \\ &= \sqrt{(\min(0.64724\text{E}+12, 0.54349\text{E}+12) / 334501.781)} \\ &= 1274.666 \text{ mm} \end{aligned}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$\begin{aligned} &= rg / L \\ &= 1274.666 / 2800.0 \\ &= 0.455 \end{aligned}$$

Location of the Skirt Centroid at critical height [e]:

$$\begin{aligned} &= \sqrt{ex^2 + ey^2} \\ &= \sqrt{(171.404^2 + -0.0^2)} \\ &= 0.000 \text{ mm} \end{aligned}$$

Where $ex = 171.404$: $ey = -0.0$

Moment Increment at Centroid [ΔM_4]:

$$\begin{aligned} &= \text{abs}(F_1 - F_f - F_g) \cdot e = F_4 \cdot e \\ &= \text{abs}(-29686.28 - 1304569 - 0.0) \cdot 171.404 \\ &= 228788752.000 \text{ N-mm} \end{aligned}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Weakening Factor, 16.12-70(b) [psi1]:

$$\begin{aligned} &= \min(1; A_4 / (\pi \cdot D_3 \cdot ea_3)) \\ &= \min(1; 334501.781 / (\pi \cdot 3789.0 \cdot 30.4)) \\ &= 0.924 \end{aligned}$$

Weakening Factor, 16.12-70(c) [psi2]:

$$\begin{aligned} &= \min(1; 4 \cdot W_4 / (\pi \cdot D_3^3 / (\pi \cdot D_3 \cdot ea_3))) \\ &= \min(1; 4.0 \cdot 0.26115\text{E}+09 / (\pi \cdot 3789.0^2 \cdot 30.4)) \\ &= 0.762 \end{aligned}$$

Maximum Bending Moment [Mmax]:

$$\begin{aligned} &= \pi / 4 \cdot d_3^2 \cdot ea_3 \cdot \sigma_{\text{call}} \\ &= \pi / 4 \cdot 3789.0^2 \cdot 30.4 \cdot 124.4 \\ &= 42644844544.0 \text{ N-mm} \end{aligned}$$

Maximum Compressive Force [Fc,max]:

$$\begin{aligned} &= \pi \cdot d_3 \cdot ea_3 \cdot \sigma_{\text{call}} \\ &= \pi \cdot 3789.0 \cdot 30.4 \cdot 124.4 \\ &= 45001380.0 \text{ N} \end{aligned}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:



$$= F4 / (\psi_1 \cdot F_{c,max}) + (\text{abs}(M_4) + F4 \cdot \text{abs}(y_g)) / (\psi_2 \cdot M_{max})$$
$$= 1334255 / (0.924 \cdot 45001380) + (\text{abs}(0.84063E+09) + 1334255 \cdot \text{abs}(171.404)) /$$
$$(0.762 \cdot 0.42645E+11)$$
$$= 0.065 \text{ must be } \leq \text{ to } 1, \text{ Passed}$$

The stresses are satisfactory in the region with the opening(s).

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Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

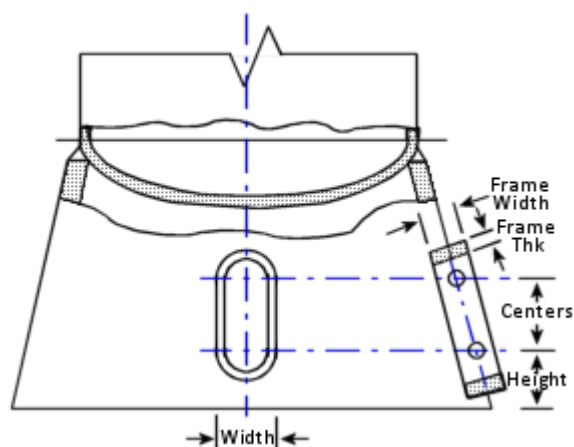
The access opening(s) is/are in section 4-4 only. The other calculations are for areas where no access openings exist. Please be aware of this when you review the results.

Item: Skirt-1

Total vessel (empty) weight	Fg	1304568.625 N
Head Weight below section 2-2	DeltaFg	197659.625 N
Weight of vessel contents (liquid)	Ff	0.000 N
Vessel Gross Weight	Fg + Ff	1304568.625 N
Skirt Mean Diameter	Dz	3789.000 mm
Skirt Analysis Thickness (Corroded)	ez	30.400 mm
Skirt Height	H	1000.000 mm
Additional Axial Force at section 1-1	F1	-29686.283 N
Moment at section 1-1	M1	593088448.000 N-mm
Applied Moment - Bottom of Skirt	Mb	840629184.000 N-mm
Weight at Bottom of Skirt	W	1316456.250 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:	Spherical
Head Mean Diameter at Skirt Top	Db 3788.000 mm
Head Corroded Thickness	eb 60.000 mm



Skirt Access Opening Input Data:

Access #	Diameter mm	Layout Angle deg	Cntr. Dist. From Bottom mm	Obround Cntr-Cntr mm	Frame Width mm	Frame Thickness mm
1	50.8000	0.0	500.00	0.00	100.000	16.000
2	50.8000	90.0	500.00	0.00	100.000	16.000
3	50.8000	180.0	500.00	0.00	100.000	16.000
4	50.8000	270.0	500.00	0.00	100.000	16.000

Interference check:

Access	Access	Pitch mm	Ligament mm
1	2	2975.8237	2925.0237
2	3	2975.8237	2925.0237
3	4	2975.8237	2925.0237
1	4	2975.8237	2925.0237



Interference Check: No Overlap Detected between Openings

Minimum Pitch between openings is: 2975.8237 mm
 Minimum Ligament between Openings is: 2925.0237 mm

Recommended minimum Ligament [IMin]:

$$= 2 \cdot \sqrt{(D \cdot t)}$$

$$= 2 \cdot \sqrt{(3789.0 \cdot 30.4)} \text{ mm}$$

$$= 678.780 \text{ mm}$$

Carefully evaluate this acceptance criteria.

Section properties at least metal area [A]:

Number of Openings At Critical Height = 4

Access No	Layout Angle 1 Deg.	Layout Angle 2 Deg.
1	-0.7682	0.7682
2	89.2318	90.7682
3	179.2318	180.7682
4	269.2318	270.7682

The points for integrating Ixx, Iyy are as follows:

FROM THETA1 Deg	TO THETA2 Deg
0.768	89.232
90.768	179.232
180.768	269.232
-89.232	-0.768

Height of Weakest Section	H	499.988	mm
Rotation of Principle Axes	a	0.674	deg.
Centroid Position in X direction	g	0.000	mm
Centroid Position in Y direction	h	-0.000	mm
Principle Moment of Inertia	Ixx	661279866880.000	mm ⁴
Principle Moment of Inertia	Iyy	661279670272.000	mm ⁴
Distance to Extreme Fiber	ybar	1909.700	mm
Distance to Extreme Fiber	xbar	1909.700	mm
Minimum Section Modulus	W4	346274400.000	mm ³
Metal Area at the Cutting Plane	A4	368488.812	mm ²

Half Opening Width Subtended Angle at Weakest Section [δ]:

$$= \sin^{-1}(\text{Largest Opening Width} / D_z)$$

$$= \sin^{-1}(50.8/3789.0)$$

$$= 0.0134 \text{ rad (0.7682 deg)}$$

Skirt Mean Radius at Section 4-4 [r]:

$$= D_z / 2$$

$$= 3789.0/2$$

$$= 1894.5 \text{ mm}$$

Because the Code uses the words 'may be' in the evaluation below, the results should be considered optional.

Check areas for openings per EN 13445 Equation 16.12-76 [A,check]:

At,i	Adelta,i	Status	lt,i <= 8*eat,i
3200.0	1057.9	...	...
3200.0	1057.9	...	...
3200.0	1057.9	...	...
3200.0	1057.9	...	...

This calculation of ratio below is per an earlier version of EN 13445.



Opening ratio at weakest section: EN 13445 equation 16.12-76 [ratio]:

$$\begin{aligned} &= \delta \cdot \sqrt{(r / e_z)} \\ &= 0.013 \cdot \sqrt{(1894.5/30.4)} \\ &= 0.106 \end{aligned}$$

Ok as ratio is < 2 - See EN 13445 16.12.76(a).

Analysis of the Section at Weakest Section Modulus [Section 4-4]:

Skirt Corroded Thickness [ez]:

$$\begin{aligned} &= ex(new) - ci - cext \\ &= 32.0 - 0.8 - 0.8 \\ &= 30.4 \text{ mm} \end{aligned}$$

Find the moment acting at the Critical Height [M4]:

$$\begin{aligned} &= (M_{bottom} - M_{top}) \cdot (H - CutHeight) / H + M_{top} \\ &= (.84063E+09 - 0.59309E+09) \cdot (1000.0 - 499.988) / 1000.0 + 0.59309E+09 \\ &= 0.71686E+09 \text{ N-mm} \end{aligned}$$

Compute the section modulus [Zxx and Zyy]:

$$\begin{aligned} Z_{xx} &= I_{xx} / y_{bar} = 0.66128E+12 / 1909.7 = 0.34627E+09 \text{ mm}^3 \\ Z_{yy} &= I_{yy} / x_{bar} = 0.66128E+12 / 1909.7 = 0.34627E+09 \text{ mm}^3 \\ Z &= \min(Z_{xx}, Z_{yy}) = \min(.34627E+09, .34627E+09) = 0.34627E+09 \text{ mm}^3 \end{aligned}$$

The balance of the calculations are per EN 13445-3, and they may be treated as academic.

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned} &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area}(A_4))} \\ &= \sqrt{(\min(0.66128E+12, 0.66128E+12) / 368488.812)} \\ &= 1339.617 \text{ mm} \end{aligned}$$

Note: rg is not included in the analysis.

Skirt Slenderness Ratio [rg/L]:

$$\begin{aligned} &= rg / L \\ &= 1339.617 / 1000.0 \\ &= 1.340 \end{aligned}$$

Location of the Skirt Centroid at critical height [e]:

$$\begin{aligned} &= \sqrt{(ex^2 + ey^2)} \\ &= \sqrt{(0.^2 + -0.0^2)} \\ &= 171.404 \text{ mm} \end{aligned}$$

Where $ex = 0$: $ey = -0.0$

Moment Increment at Centroid [ΔM_4]:

$$\begin{aligned} &= \text{abs}(F_1 - F_f - F_g) \cdot e = F_4 \cdot e \\ &= \text{abs}(-29686.28 - 1304569 - 0.0) \cdot 0.0 \\ &= 0.000 \text{ N-mm} \end{aligned}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$\begin{aligned} &= F_1 - F_g - F_f + 4 \cdot M_1 / D_z \\ &= -29686.283 - 1304569 - 0.0 + 4 \cdot 0.59309E+09 / 3789.0 \\ &= -708392.6 \text{ N} \end{aligned}$$

Force at the joint, 16.12-2 [Fzq]:

$$\begin{aligned} &= F_1 - F_g - F_f - 4 \cdot M_1 / D_z \\ &= -29686.283 - 1304569 - 0.0 - 4 \cdot 0.59309E+09 / 3789.0 \\ &= -1960117.1 \text{ N} \end{aligned}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$\begin{aligned} &= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b) \\ &= (-0.7E+06 + 197659.625 + 0.0) / (\pi \cdot 3788.0 \cdot 60.0) + 6.416 \cdot 3788.0 / (4 \cdot 60.0) \\ &= 100.56 \text{ MPa} \end{aligned}$$

Stress [σ_{1qm}]:

$$= (F_{zq} + \Delta F_q + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-0.2E+07 + 197659.625 + 0.0) / (\pi \cdot 3788.0 \cdot 60.0) + 6.416 \cdot 3788.0 / (4 \cdot 60.0)$$

$$= 98.80 \text{ MPa}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (F_f + \Delta F_q) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (0.0 + 197659.625) / (\pi \cdot 3788.0 \cdot 60.0) + 6.416 \cdot 3788.0 / (4 \cdot 60.0)$$

$$= 101.55 \text{ MPa}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= F_{zp} / (\pi \cdot D_z \cdot e_z)$$

$$= -708392.562 / (\pi \cdot 3789.0 \cdot 30.4)$$

$$= -1.96 \text{ MPa}$$

Stress [σ_{3qm}]:

$$= F_{zq} / (\pi \cdot D_z \cdot e_z)$$

$$= -1960117.125 / (\pi \cdot 3789.0 \cdot 30.4)$$

$$= -5.42 \text{ MPa}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b)$$

$$= 1 - 0.5(3788.0 + 60.0 - 3789.0 + 30.4) / (1864.0 + 60.0)$$

$$= 0.9768$$

$\gamma = \arccos(\cos(\gamma)) = \arccos(0.977) = 12.375 \text{ deg}$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(e_b^2 + e_z^2 + 2 \cdot e_b \cdot e_z \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(60.0^2 + 30.4^2 + 2 \cdot 60.0 \cdot 30.4 \cdot 0.977)}$$

$$= 44.965 \text{ mm}$$

Bending moment at verification points p and q [Mp & Mq]:

$$M_p = a \cdot F_{zp} = 44.96 \cdot -708392.6 = -31.8658 \times 10^6 \text{ N-mm}$$

$$M_q = a \cdot F_{zq} = 44.96 \cdot -1960117.1 = -88.1724 \times 10^6 \text{ N-mm}$$

Correction factor [C]:

$$= 0.63 - 0.057(e_b/e_z)^2$$

$$= 0.63 - 0.057(60.0/30.4)^2$$

$$= 0.408$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot e_b^2)$$

$$= 0.408 \cdot 6 \cdot -0.3E+08 / (\pi \cdot 3788.0 \cdot 60.0^2)$$

$$= -1.82 \text{ MPa}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot e_b^2)$$

$$= 0.408 \cdot 6 \cdot -0.9E+08 / (\pi \cdot 3788.0 \cdot 60.0^2)$$

$$= -5.04 \text{ MPa}$$

Stress [$\sigma_{3pb(o)}$]:

$$= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot e_z^2)$$

$$= 0.408 \cdot 6 \cdot -0.3E+08 / (\pi \cdot 3789.0 \cdot 30.4^2)$$

$$= -7.09 \text{ MPa}$$

Stress [$\sigma_{3qb(o)}$]:

$$= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot e_z^2)$$

$$= 0.408 \cdot 6 \cdot -0.9E+08 / (\pi \cdot 3789.0 \cdot 30.4^2)$$

$$= -19.61 \text{ MPa}$$



Geometry variables, 16.12-32 to 36:

$$eb/Db = 2.362/149.134 = 0.0158$$

$$y = 125 * eb/Db = 125 * 0.0158 = 1.9799$$

$$\alpha = 0$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$= 0.0 \text{ MPa (16.12-31)}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)}$$

$$= 100.556 - -1.82 + 0.0$$

$$= 102.38 \text{ MPa}$$

Stress [σ_{1potot}]:

$$= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)}$$

$$= 100.556 + -1.82 - 0.0$$

$$= 98.74 \text{ MPa}$$

Stress [σ_{1qitot}]:

$$= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)}$$

$$= 98.803 - -5.036 + 0.0$$

$$= 103.84 \text{ MPa}$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 98.803 + -5.036 - 0.0$$

$$= 93.77 \text{ MPa}$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 101.549 + -1.82 + 0.0$$

$$= 99.73 \text{ MPa}$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 101.549 - -1.82 - 0.0$$

$$= 103.37 \text{ MPa}$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 101.549 + -5.036 + 0.0$$

$$= 96.51 \text{ MPa}$$

Stress [σ_{2qotot}]:

$$= \sigma_{2qm} - \sigma_{2qb(o)} - \sigma_{2b(p)}$$

$$= 101.549 - -5.036 - 0.0$$

$$= 106.58 \text{ MPa}$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= -1.958 - -7.088$$

$$= 5.13 \text{ MPa}$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= -1.958 + -7.088$$

$$= -9.05 \text{ MPa}$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -5.417 - -19.613$$

$$= 14.20 \text{ MPa}$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -5.417 + -19.613$$

= -25.03 MPa

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{1pm} / f)^2)$$

$$= 101.028(3 - (1/1.5) \cdot (100.556/101.028)^2)$$

$$= 236.36 \text{ MPa}$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{1qm} / f)^2)$$

$$= 101.028(3 - (1/1.5) \cdot (98.803/101.028)^2)$$

$$= 238.66 \text{ MPa}$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{2pm} / f)^2)$$

$$= 101.028(3 - (1/1.5) \cdot (101.549/101.028)^2)$$

$$= 235.03 \text{ MPa}$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{2qm} / f)^2)$$

$$= 101.028(3 - (1/1.5) \cdot (101.549/101.028)^2)$$

$$= 235.03 \text{ MPa}$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 101.028(3 - (1/1.5) \cdot (-1.958/101.028)^2)$$

$$= 303.06 \text{ MPa}$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 101.028(3 - (1/1.5) \cdot (-5.417/101.028)^2)$$

$$= 302.89 \text{ MPa}$$

Check of the Skirt in Regions with Openings, per 16.12.4.3:

Weakening Factor, 16.12-70(b) [psi1]:

$$= \min(1; A4 / (\pi \cdot D3 \cdot ea3))$$

$$= \min(1; 368488.812 / (\pi \cdot 3789.0 \cdot 30.4))$$

$$= 1.000$$

Weakening Factor, 16.12-70(c) [psi2]:

$$= \min(1; 4 \cdot W4 / (\pi \cdot D^3 / (\pi \cdot D3 \cdot ea3)))$$

$$= \min(1; 4.0 \cdot 0.34627E+09 / (\pi \cdot 3789.0^2 \cdot 30.4))$$

$$= 1.000$$

Maximum Bending Moment [Mmax]:

$$= \pi/4 \cdot d3^2 \cdot ea3 \cdot \sigma_{call}$$

$$= \pi/4 \cdot 3789.0^2 \cdot 30.4 \cdot 124.4$$

$$= 42644844544.0 \text{ N-mm}$$

Maximum Compressive Force [Fc,max]:

$$= \pi \cdot d3 \cdot ea3 \cdot \sigma_{call}$$

$$= \pi \cdot 3789.0 \cdot 30.4 \cdot 124.4$$

$$= 45001380.0 \text{ N}$$

Unity Check of Weakened Section, 16.12-70 [UC1]:

$$= F4 / (\psi_{i1} \cdot F_{c,max}) + (abs(M4) + F4 \cdot abs(yq)) / (\psi_{i2} \cdot M_{max})$$

$$= 1334255 / (1.0 \cdot 45001380) + (abs(0.71686E+09) + 1334255 \cdot abs(0.0)) / (1.0 \cdot 0.42645E+11)$$

$$= 0.046 \text{ must be } \leq \text{ to } 1, \text{ Passed}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	102.38	236.36	Pass	46



Stot1po	98.74	236.36	Pass	47
Stot1qi	103.84	238.66	Pass	48
Stot1qo	93.77	238.66	Pass	49
Stot2pi	99.73	235.03	Pass	50
Stot2po	103.37	235.03	Pass	51
Stot2qi	96.51	235.03	Pass	52
Stot2qo	106.58	235.03	Pass	53
Stot3pi	5.13	303.06	Pass	54
Stot3po	-9.05	303.06	Pass	55
Stot3qi	14.20	302.89	Pass	56
Stot3qo	-25.03	302.89	Pass	57

The stresses are satisfactory in the region without openings.
The stresses are satisfactory in the region with the opening(s).

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
Packing is installed in the Shop.
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	5757.637 mm
Center of Gravity of the Packing	5287.500 mm
Center of Gravity of the Insulation	5292.728 mm
Center of Gravity of the Nozzles	5162.120 mm
Center of Gravity of the Added Weights (Operating)	5434.234 mm
Center of Gravity of the Added Weights (Empty)	4943.981 mm
Center of Gravity of Bare Shell New and Cold	4691.313 mm
Center of Gravity of Bare Shell Corroded	4698.394 mm
Vessel CG in the Operating Condition	5016.710 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4947.981 mm
Vessel CG in the Test Condition	5049.121 mm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	3821.0000	mm
Skirt Thickness	STHK	32.0000	mm
Skirt Internal Corrosion Allowance	SCA	0.8000	mm
Skirt External Corrosion Allowance		0.8000	mm
Skirt Material		SA-516 70	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	32.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-36	
Basing Ope. All. Stress	BASOPE, f4	114.00	MPa
Basing Yield Stress		248.00	MPa
Inside Diameter of Basing	DI, D4	3677.0000	mm
Outside Diameter of Basing	DOU, D5	4077.0000	mm

Nominal Diameter of Bolts	BND, d(b0)	36.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-307 A	
Bolt Material Class/Thickness Range		6 <= t <= 100	
Bolt Operating Allowable Stress	SA, fB	138.00	MPa
Number of Bolts	RN, nb	24	
Diameter of Bolt Circle	DC, DBC	3949.0000	mm
Bolt Allowable Shear Stress		82.800	MPa

Ultimate Comp. Strength of Concrete	FPC	41.0	MPa
Allowable Comp. Strength of Concrete	FC	22.5	MPa
Modular ratio Steel/Concrete		9.833	

Thickness of Gusset Plates	TGA, e7	16.0000	mm
Width of Gussets at Top Plate	TWDT	118.0000	mm
Width of Gussets at Base Plate	BWDT	118.0000	mm
Gusset Plate Elastic Modulus	E	201745.7	MPa
Gusset Plate Yield Stress	SY	248.0	MPa
Height of Chair	h1	257.0000	mm
Distance between Gussets	RG, b6	76.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	38.0000	mm
Number of Gussets per bolt	NG	2	

Thickness of Top Plate or Ring	TTA, e5	32.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	128.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	50.0000	mm

External Corrosion Allowance	CA	3.0000	mm
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Dead Weight of Vessel	DW	1376441.8	N
Operating Weight of Vessel	ROW	1455380.0	N
Earthquake Moment on Basing	EQMOM	1256478080.0	N-mm
Wind Moment on Basing	WIMOM	127071056.0	N-mm

User Moment at the Base [Earthquake Case]		288874432.0	N-mm (UEQMOM)
User Moment at the Base [WindLoad Case]		288874432.0	N-mm (UWIMOM)
Percent Bolt Preload	ppl	100.0	

Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	

Factor for Increase of Allowables	Fact	1.0000	
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Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$



$$= ((4 \cdot 0.15454E+10/3949.0) - 1410670) / 24$$

$$= 6416.9 \text{ N}$$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.41595E+09/3949.0) - 1376442) / 24$$

$$= -39804.0 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/3949.0) - 1455380) / 24$$

$$= -60640.8 \text{ N}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded)	Abss :	600.6298	mm ²
Area Available in all the Bolts	Abss * RN :	14415.1162	mm ²



*There seems to be no tensile load on bolts per the Brownell and Young Method.
It is possible that this method is encountering a problem. It is strongly
recommended that you try the simplified analysis method for the basing.*

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 1.082[(2 \cdot 1.0 \cdot 3949.0 + 200.0) / (2 \cdot 1.0 \cdot 3949.0)]$$

$$= 1.109 \text{ MPa, Should be } \leq 22.5 \text{ MPa}$$

Values for table 10.3, l = 128.0, b = 258.461, l/b = 0.495238

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(2125.068, 5852.549) = 5852.549 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 5852.549 / 163.7)} + 3.0$$

$$= 17.648 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 128.0, b = 76.0, b/l = 0.59375

As b/l (0.594) is less than 1, inverting b/l = 1.684.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2 \sin(\pi \cdot a/l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4\pi)]$$

$$= 82879.86 / (4 \cdot \pi) [1.3(\ln((2 \cdot 128.0 \cdot \sin(\pi \cdot 64.0/128.0)) / (\pi \cdot 30.0))) + 1] - [(0.7 - 0.051) \cdot 82879.86 / (4 \cdot \pi)]$$

$$= 10882.5547 \text{ N}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot Abs(Mo) / Fb)} + Ca$$

$$= \sqrt{(6 \cdot Abs(10882.55) / 165.33)} + 3.0$$

$$= 22.8739 \text{ mm}$$

Required Gusset Plate Thickness [tg]:

$$= P / (\text{Stress Term} \cdot lavg) + Ca$$

$$= 82879.86 / (124.11 \cdot 118.0) + 3.0$$

$$= 12.525 \text{ (not less than } 9.525 + 3.0) \text{ mm}$$

Bolt spacing [m]:



$$= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$$

$$= \pi \cdot 3949.0 / 24 = 516.923 \text{ mm}$$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:

$$= 1.76 \left(P \cdot a / (m \cdot (h + tba)) \cdot 1.5 \cdot Sktope \right)^{2/3} \cdot r^{1/3} + Ca$$

$$= 1.76 (82880 \cdot 64.0 / (516.92 \cdot 225.0 \cdot 207))^{2/3} \cdot 1910.51^{1/3} + Ca$$

$$= 7.967 + 1.6 = 9.567 \text{ mm}$$

Shear Stress in a Single Bolt [tb]:

$$= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 253869.6 / (600.63 \cdot 24)$$

$$= 17.6 \text{ MPa. Must be less than 82.8 MPa.}$$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	17.6478	mm
Actual Basing Thickness as entered by user	32.0000	mm
Required Thickness of Chair Cap	22.8739	mm
Actual Top Ring Thickness as entered by user	32.0000	mm
Required Gusset thickness, + CA	12.5250	mm
Actual Gusset Thickness as entered by user	16.0000	mm
Required Thickness of Skirt for Local Stress	9.5669	mm
Given Thickness of Skirt	32.0000	mm
Required Gusset Height to meet local stress	0.3984	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:

$$= \text{SkirtStress} \cdot (\text{SkirtThickness} - CA)$$

$$= 8.569 \cdot (32.0 - 1.6)$$

$$= 260.46 \text{ N/mm}$$

Results for Computed Minimum Basing Weld Size [BWeld]:

$$= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 260 / [(0.4 \cdot 248) \cdot 2 \cdot 0.707]$$

$$= 1.857 \text{ mm}$$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:

$$= \text{Bolt Load} / (Cmwth + 2 \cdot (Hg + Tta))$$

$$= 82879.9 / (180.0 + 2 \cdot (193.0 + 32.0))$$

$$= 131.555 \text{ N/mm}$$

Horizontal Plate Load [Wh]:

$$= \text{Bolt Load} \cdot e / (Cmwth \cdot (Hg + Tta) + 0.6667 \cdot (Hg + Tta)^2)$$

$$= 82879.9 \cdot 64.0 / (180.0 \cdot (225.0) + 0.6667 \cdot (225.0)^2)$$

$$= 71.438 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{131.56^2 + 71.44^2}$$

$$= 149.701 \text{ N/mm}$$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 149.7 / [(0.4 \cdot 248) \cdot 2 \cdot 0.707]$$

$$= 1.067 \text{ mm}$$

Results for Computed Minimum Gusset to Top Plate Weld Size

Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 82879.9 / (2 \cdot 118.0)$$

$$= 351.186 \text{ N/mm}$$

Weld Load [Wh]:

$$\begin{aligned} &= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth}) \\ &= 82879.9 \cdot 64.0 / (2 \cdot 225.0 \cdot 118.0) \\ &= 99.893 \text{ N/mm} \end{aligned}$$

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{ W_v^2 + W_h^2 } \\ &= \sqrt{ 351.19^2 + 99.89^2 } \\ &= 365.117 \text{ N/mm} \end{aligned}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$\begin{aligned} &= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 365.12 / [(0.4 \cdot 248) \cdot 2 \cdot 0.707] \\ &= 2.603 \text{ mm} \end{aligned}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm
Required Gusset to Skirt Double Fillet Weld Size	6.3500	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	2.6032	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N2400

From: 30

Pressure for Reinforcement Calculations	P	6.4164	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Hemispherical Head	D	3722.00	mm
Head Finished (Minimum) Thickness	t	63.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	88.90	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		400.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

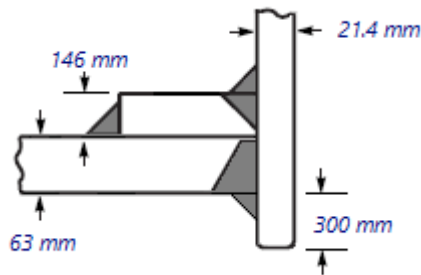
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	62.0000	mm
Inside Projection	h	300.0000	mm
Weld leg size, Inside Element to Shell	Wi	14.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	101.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	723.8500	mm
Thickness of Pad	te	146.0000	mm
Weld leg size between Pad and Shell	Wp	110.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	146.0000	mm
Reinforcing Pad Width		158.7250	mm

Flange Class		600	
Flange Grade		GR 1.1	

Piping Exit Angle [North Clockwise reference]		90.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		2500.0	mm
Centerline Distance to Tangent		2800.0	mm

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2400

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	406.400 mm.
Actual Thickness Used in Calculation	21.412 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 6.42 \cdot 1864.0 / (2 \cdot 101 \cdot 1.0 - 0.2 \cdot 6.42)$
 $= 59.5707 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.42 \cdot 203.2 / (89 \cdot 1.0 + 0.4 \cdot 6.42)$
 $= 14.2545 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0528 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	739.1512 mm
Parallel to Vessel Wall, opening length	d	369.5756 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		150.0000 mm
Normal to Vessel Wall, Inward		38.5305 mm

Weld Strength Reduction Factor [fr1]:

$= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101.0)$
 $= 0.880$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101.0)$
 $= 0.880$

Weld Strength Reduction Factor [fr4]:

$= \min(1, S_p / S_v)$
 $= \min(1, 101.0 / 101.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$
 $= \min(0.88, 1.)$
 $= 0.880$

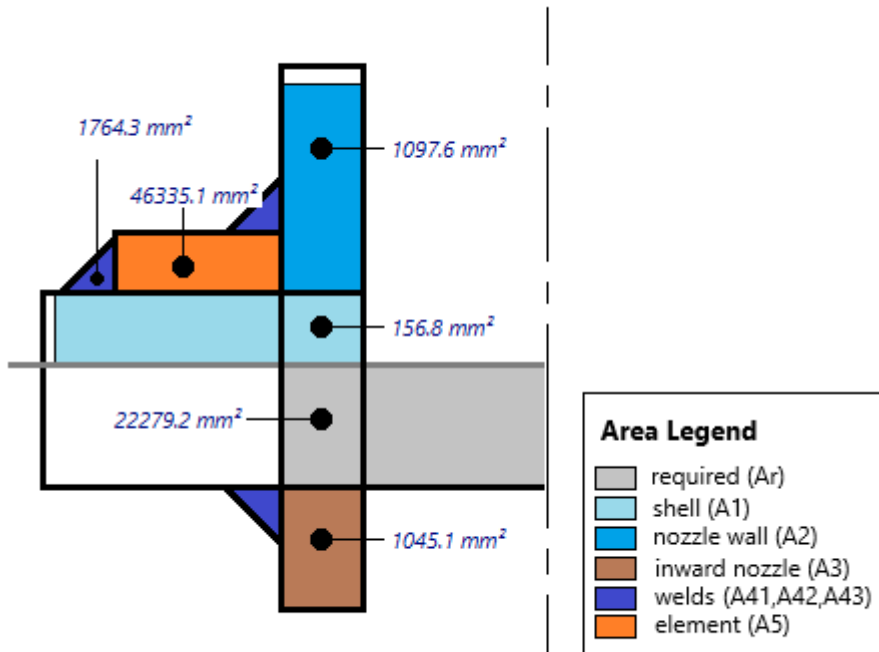


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	22279.207	1146.954	NA
Area in Shell	A1	156.765	19669.625	NA
Area in Nozzle Wall	A2	1097.590	4582.663	NA
Area in Inward Nozzle	A3	1045.108	1045.108	NA
Area in Welds A41+A42+A43		1764.260	1764.260	NA
Area in Element	A5	46335.051	46335.051	NA
TOTAL AREA AVAILABLE	Atot	50398.773	73396.703	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 531.1976 146.0000 mm
 Based on given Pad Diameter: 723.8500 57.3963 mm
 Based on the Estimated Diameter Limit: 737.5637 55.0195 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (369.5756 \cdot 59.5707 \cdot 1.0 + 2 \cdot 18.4122 \cdot 59.5707 \cdot 1.0 \cdot (1 - 0.88))$
 $= 22279.207 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 369.576(1.0 \cdot 60.0 - 1.0 \cdot 59.571) - 2 \cdot 18.412$
 $(1.0 \cdot 60.0 - 1.0 \cdot 59.5707) \cdot (1 - 0.88)$
 $= 156.765 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 150.0) \cdot (18.41 - 14.25) \cdot 0.88$

= 1097.590 mm²

Area Available in Inward Nozzle [A3]:

= $2 \cdot t_i \cdot \min(h, Tl, 2.5 \cdot t_i) \cdot fr2$
= $2 \cdot 15.4122 \cdot (38.5305) \cdot 0.88$
= 1045.108 mm²

Area Available in Welds [A41 + A42 + A43]:

= $(Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can/0.707)^2 - Ar \text{ Lost}) \cdot fr2 + \text{Trapfr4}$
= $(64.0) \cdot 0.88 + (95.2053) \cdot 0.88 + 1624.6099^2 \cdot 1$
= 1764.260 mm²

Area Available in Element [A5]:

= $(\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
= $(723.85 - 406.4) \cdot 146.0 \cdot 0.9997$
= 46335.051 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 17.2545 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 62.5707 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 62.5707 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.9542 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]
= min[11.331, max(62.5707, 4.5)]
= 11.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(17.2545, 11.3312)
= 17.2545 mm

Available Nozzle Neck Thickness = $0.875 \cdot 21.412 = 18.736$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 54.0,	Allowable	: 88.9 MPa	Passed
Expansion	: 0.0,	Allowable	: 204.6 MPa	Passed
Occasional	: 30.7,	Allowable	: 118.2 MPa	Passed
Shear	: 12.8,	Allowable	: 62.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 18.736, tr = 10.814, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.687$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	10 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-7 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 18.736, tr = 10.814, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.687$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 18.736, tr = 10.814, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.687$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -27 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $tg = 63.0$, $tr = 43.537$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.726$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 22 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 7 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 18.736$, $tr = 10.814$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.687$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -27 °C

Gov. MDMT of the Nozzle : -7 °C
 Gov. MDMT of the Reinforcement Pad : 7 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 7 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.42/10.20 = 0.629$

Weld Size Calculations, Description: N2400

Intermediate Calc. for nozzle/shell Welds	Tmin	18.4122	mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000	mm
Intermediate Calc. for Inward Weld	TminIns	18.4122	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 * T_{minPad}$	$77.7700 = 0.7 * W_p \text{ mm}$
Inward Weld	$6.0000 = \text{Min per Code}$	$6.8980 = 0.7 * W_i\text{-Can mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (22279.207 - 156.7648 + 2 \cdot 18.4122 \cdot 0.88 \cdot (1.0 \cdot 60.0 - 59.5707)) \cdot 101)$
 $= 2236192.25 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i\text{-Can} / .707)^2 \cdot fr2) \cdot Sv$
 $= (1097.5896 + 46335.0508 + 1764.2603 - 95.2053 \cdot 0.88) \cdot 101$
 $= 4961358.50 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (1097.5896 + 1045.1079 + 171.7724 + (1944.2352)) \cdot 101$
 $= 430210.06 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (1097.5896 + 1045.1079 + 1764.2603 + 46335.0508 + (1944.2352)) \cdot 101$
 $= 5271802.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 406.4 \cdot 10.0 \cdot 0.49 \cdot 89 \\ &= 278057. \text{ N} \end{aligned}$$

Shear, Inward Nozzle Weld [Sinw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 406.4 \cdot 9.7573 \cdot 0.49 \cdot 89 \\ &= 271310. \text{ N} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW} \\ &= (3.1416/2.0) \cdot 723.85 \cdot 110.0 \cdot 0.49 \cdot 101 \\ &= 6189302. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 193.9939) \cdot (21.4122 - 3.0) \cdot 0.7 \cdot 89 \\ &= 698243. \text{ N} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg} \\ &= (3.1416/2) \cdot 406.4 \cdot 146.0 \cdot 0.74 \cdot 101 \\ &= 6965344. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 406.4 \cdot (63.0 - 3.0) \cdot 0.74 \cdot 101 \\ &= 2863252. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (6189303 + 698243) = 6887546 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (278057 + 6965345 + 2863252 + 271310) = 10377963 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (6189303 + 2863252 + 271310) = 9323864 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 6887545 N	, must exceed W = 2236192 N	or W1 = 4961358 N
Path 2-2 = 10377963N	, must exceed W = 2236192 N	or W2 = 430210 N
Path 3-3 = 9323864 N	, must exceed W = 2236192 N	or W3 = 5271802 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 11.1268 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 563.0000 mm

Input Echo, WRC107/537 Item 1, Description: N2400 :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	3.0000 mm
Vessel Diameter	Dv	3722.000 mm
Vessel Thickness	Tv	63.000 mm



Design Temperature	T1	400.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	101.03	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	406.400	mm
Nozzle Thickness	Tn	21.412	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	88.90	MPa
Thickness of Reinforcing Pad	Tpad	146.000	mm
Diameter of Reinforcing Pad	Dpad	723.850	mm
Design Internal Pressure	Dp	6.416	MPa
Include Pressure Thrust		No	

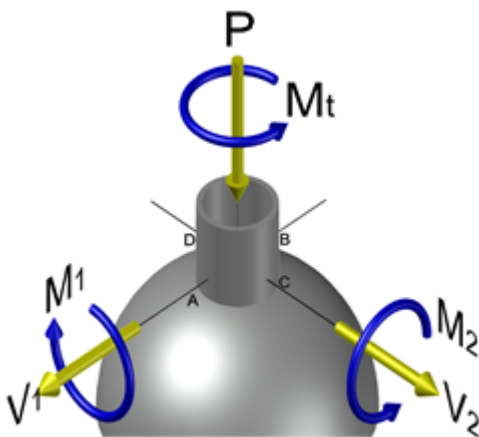
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	36000.0	N
Longitudinal Shear (SUS)	(V1) V1	26000.0	N
Circumferential Shear (SUS)	(Vc) V2	26000.0	N
Circumferential Moment (SUS)	(Mc) M1	32000000.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	32000000.0	N-mm
Torsional Moment (SUS)	Mt	36799944.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 406.4 + 2 \cdot 1.65 \cdot \sqrt{(1894.0 (63.0 - 3.0))}$$

$$= 1518.848 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	36000.0	N
Circumferential Shear	(VC) V2	26000.0	N
Longitudinal Shear	(VL) V1	26000.0	N
Circumferential Moment	(MC) M1	32000000.0	N-mm
Longitudinal Moment	(ML) M2	32000000.0	N-mm
Torsional Moment	MT	36799944.0	N-mm

Dimensionless Param: U = 0.32 TAU = 10.54 RHO = 4.00 (11.19)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 4	0.06127	(A,B,C,D)
M(x) / P	SP 4	0.02540	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 4	0.18480	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 4	0.08284	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 4	0.18480	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 4	0.08284	(A,B,C,D)
N(y) * T / P	SP 4	0.34219	(A,B,C,D)
M(y) / P	SP 4	0.12973	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 4	0.26780	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 4	0.48880	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 4	0.26780	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 4	0.48880	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Rad.	Bend. P	-0.1	0.1	-0.1	0.1	-0.1	0.1	-0.1	0.1
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-0.6	0.6	0.6	-0.6
Rad.	Memb. ML	-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-0.6	0.6	0.6	-0.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-1.0	0.4	0.6	-0.3	-1.0	0.4	0.6	-0.3
Tang.	Memb. P	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3
Tang.	Bend. P	-0.7	0.7	-0.7	0.7	-0.7	0.7	-0.7	0.7
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-3.5	3.5	3.5	-3.5
Tang.	Memb. ML	-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-3.5	3.5	3.5	-3.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-4.7	3.5	2.8	-2.8	-4.7	3.5	2.8	-2.8
Shear VC		0.2	0.2	-0.2	-0.2	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear MT		0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Tot. Shear		0.9	0.9	0.5	0.5	0.5	0.5	0.9	0.9
Str. Int.		4.9	3.8	2.9	2.9	4.8	3.6	3.2	3.1

Unitless Prm: U = 1.07 TAU = 0.00 (19.16) RHO = 0.00 (3.26)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
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N(y) * T / P	SR 2	0.01820	(A, B, C, D)
M(y) / P	SR 2	0.00957	(A, B, C, D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.02046	(A, B, C, D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.01871	(A, B, C, D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.02046	(A, B, C, D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.01871	(A, B, C, D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6	-0.6
Rad.	Bend. P	-1.9	1.9	-1.9	1.9	-1.9	1.9	-1.9	1.9
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-1.8	-1.8	1.8	1.8
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-9.8	9.8	9.8	-9.8
Rad.	Memb. ML	-1.8	-1.8	1.8	1.8	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-9.8	9.8	9.8	-9.8	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-14.1	9.2	9.1	-6.7	-14.1	9.2	9.1	-6.7
Tang. Memb. P		-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2	-0.2
Tang. Bend. P		-0.6	0.6	-0.6	0.6	-0.6	0.6	-0.6	0.6
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Tang. Bend. MC		0.0	0.0	0.0	0.0	-3.0	3.0	3.0	-3.0
Tang. Memb. ML		-0.5	-0.5	0.5	0.5	0.0	0.0	0.0	0.0
Tang. Bend. ML		-3.0	3.0	3.0	-3.0	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-4.3	2.8	2.7	-2.0	-4.3	2.8	2.7	-2.0
Shear VC		0.4	0.4	-0.4	-0.4	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Shear MT		0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Tot. Shear		1.1	1.1	0.4	0.4	0.4	0.4	1.1	1.1
Str. Int.		14.2	9.4	9.2	6.7	14.1	9.3	9.3	7.0

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress Load		Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	29.3	29.3	29.3	29.3	29.3	29.3	29.3	29.3
Rad.	Pl (SUS)	-0.3	-0.3	0.2	0.2	-0.3	-0.3	0.2	0.2
Rad.	Q (SUS)	-0.7	0.7	0.5	-0.5	-0.7	0.7	0.5	-0.5
Long.	Pm (SUS)	29.3	29.3	29.3	29.3	29.3	29.3	29.3	29.3
Long.	Pl (SUS)	-0.6	-0.6	0.0	0.0	-0.6	-0.6	0.0	0.0
Long.	Q (SUS)	-4.1	4.1	2.8	-2.8	-4.1	4.1	2.8	-2.8
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.2	0.2	-0.2	-0.2	-0.2	-0.2	0.2	0.2
Shear	Q (SUS)	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7



Pm (SUS)	29.3	29.3	29.3	29.3	29.3	29.3	29.3	29.3
Pm+Pl (SUS)	29.1	29.1	29.6	29.6	29.1	29.1	29.6	29.6
Pm+Pl+Q (Total)	28.5	33.0	32.2	29.1	28.3	32.9	32.4	29.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	29.25	101.03	Passed
Pm+Pl (SUS)	29.56	151.54	Passed
Pm+Pl+Q (TOTAL)	33.02	303.08	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	99.7	99.7	99.7	99.7	99.7	99.7	99.7	99.7
Rad.	Pl (SUS)	-2.4	-2.4	1.2	1.2	-2.4	-2.4	1.2	1.2
Rad.	Q (SUS)	-11.7	11.7	7.9	-7.9	-11.7	11.7	7.9	-7.9
Long.	Pm (SUS)	99.7	99.7	99.7	99.7	99.7	99.7	99.7	99.7
Long.	Pl (SUS)	-0.7	-0.7	0.4	0.4	-0.7	-0.7	0.4	0.4
Long.	Q (SUS)	-3.5	3.5	2.4	-2.4	-3.5	3.5	2.4	-2.4
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.4	0.4	-0.4	-0.4	-0.4	-0.4	0.4	0.4
Shear	Q (SUS)	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Pm (SUS)		99.7	99.7	99.7	99.7	99.7	99.7	99.7	99.7
Pm+Pl (SUS)		99.1	99.1	101.1	101.1	99.1	99.1	101.1	101.1
Pm+Pl+Q (Total)		95.6	109.2	108.9	97.7	95.5	109.0	109.1	98.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	99.74	101.03	Passed
Pm+Pl (SUS)	101.10	151.54	Passed
Pm+Pl+Q (TOTAL)	109.18	303.08	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: U1600

From: 30

Pressure for Reinforcement Calculations	P	6.4164	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Hemispherical Head	D	3722.00	mm
Head Finished (Minimum) Thickness	t	63.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1500.0000	mm
-------------------------------	----	-----------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+IT
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	101.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		600.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	22.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

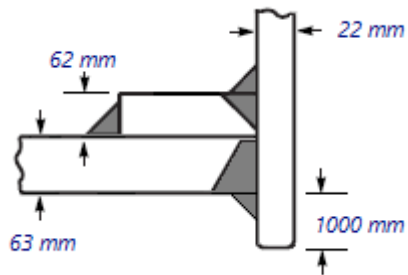
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	1500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	14.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	62.0000	mm
Inside Projection	h	1000.0000	mm
Weld leg size, Inside Element to Shell	Wi	14.0000	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	101.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1120.0000	mm
Thickness of Pad	te	62.0000	mm
Weld leg size between Pad and Shell	Wp	62.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	62.0000	mm
Reinforcing Pad Width		260.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: U1600

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	600.000 mm.
Actual Thickness Used in Calculation	22.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 6.42 \cdot 1864.0 / (2 \cdot 101 \cdot 1.0 - 0.2 \cdot 6.42)$
 $= 59.5707 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.42 \cdot 300.0 / (101 \cdot 1.0 + 0.4 \cdot 6.42)$
 $= 18.5862 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 3.0634 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	DI	1124.0001 mm
Parallel to Vessel Wall, opening length	d	562.0001 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		109.5000 mm
Normal to Vessel Wall, Inward		40.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 101.0 / 101.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 101.0 / 101.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 101.0 / 101.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.)$
 $= 1.000$

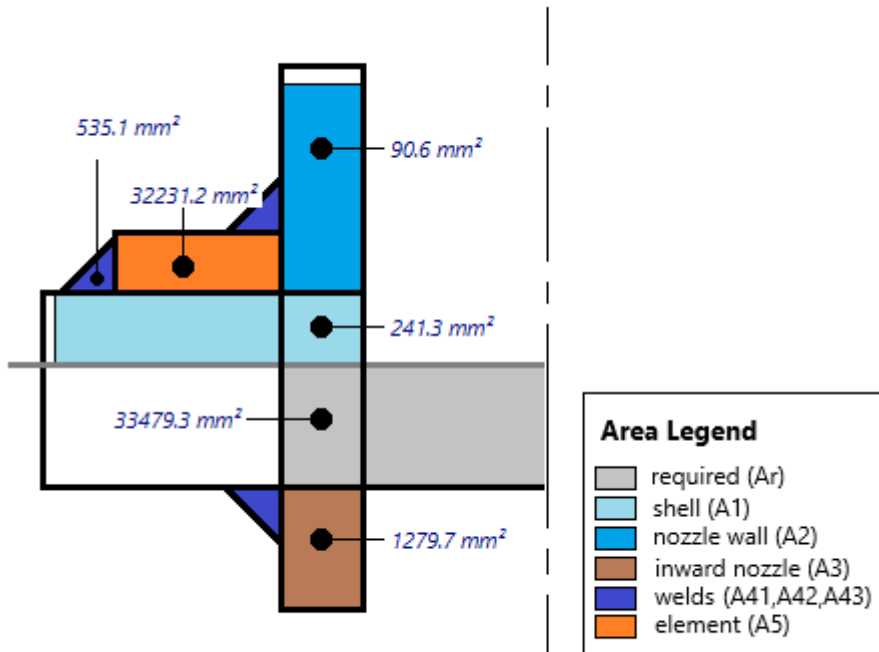


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	33479.348	1723.548	NA
Area in Shell	A1	241.268	30272.410	NA
Area in Nozzle Wall	A2	90.591	3489.153	NA
Area in Inward Nozzle	A3	1279.651	1279.651	NA
Area in Welds	A41+A42+A43	535.063	535.063	NA
Area in Element	A5	32231.203	32231.203	NA
TOTAL AREA AVAILABLE	Atot	34377.773	67807.477	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1105.5054 62.0000 mm
 Based on given Pad Diameter: 1120.0000 60.2718 mm
 Based on the Estimated Diameter Limit: 1122.4126 59.9934 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (562.0 \cdot 59.5707 \cdot 1.0 + 2 \cdot 19.0 \cdot 59.5707 \cdot 1.0 \cdot (1 - 1.))$
 $= 33479.348 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 562.0(1.0 \cdot 60.0 - 1.0 \cdot 59.571) - 2 \cdot 19.0$
 $(1.0 \cdot 60.0 - 1.0 \cdot 59.5707) \cdot (1 - 1.)$
 $= 241.268 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 109.5) \cdot (19.0 - 18.59) \cdot 0.9997$

= 90.591 mm²

Area Available in Inward Nozzle [A3]:

$$\begin{aligned} &= 2 \cdot t_i \cdot \min(h, Tl, 2.5 \cdot t_i) \cdot fr2 \\ &= 2 \cdot 16.0 \cdot (40.0) \cdot 0.9997 \\ &= 1279.651 \text{ mm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= (W_o^2 - Ar \text{ Lost}) \cdot Fr3 + ((W_i - can/0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4 \\ &= (196.0) \cdot 1. + (95.2053) \cdot 1. + 244.0035^2 \cdot 1. \\ &= 535.063 \text{ mm}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (1120.0 - 600.0) \cdot 62.0 \cdot 0.9997 \\ &= 32231.203 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 21.5862 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 62.5707 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 62.5707 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.9542 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[11.331, \max(62.5707, 4.5)] \\ &= 11.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(21.5862, 11.3312) \\ &= 21.5862 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 22.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 22.0, tr = 13.694, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.721, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	15 °C
Min Metal Temp. at Required thickness (UCS 66.1)	0 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 62.0, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.726, Temp. Reduction = 15 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	6 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 22.0, tr = 13.694, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E \cdot / (tg - c) = 0.721$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -20 °C

Gov. MDMT of the Nozzle : 0 °C
 Gov. MDMT of the Reinforcement Pad : 6 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 6 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.42/10.20 = 0.629$

Weld Size Calculations, Description: U1600

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm
 Intermediate Calc. for Inward Weld TminIns 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	9.8980 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	43.8340 = 0.7 * Wp mm
Inward Weld	6.0000 = Min per Code	6.8980 = 0.7 * Wi-Can mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (33479.3477 - 241.2678 + 2 \cdot 19.0 \cdot 0.9997 \cdot (1.0 \cdot 60.0 - 59.5707)) \cdot 101)$
 $= 3359325.00 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (90.5907 + 32231.2031 + 535.0627 - 95.2053 \cdot 1.) \cdot 101$
 $= 3309551.25 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (90.5907 + 1279.6506 + 291.1258 + (2279.3774)) \cdot 101$
 $= 398090.03 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (90.5907 + 1279.6506 + 535.0627 + 32231.2031 + (2279.3774)) \cdot 101$
 $= 3678695.75 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 600.0 \cdot 14.0 \cdot 0.49 \cdot 101$
 $= 652950. \text{ N}$

Shear, Inward Nozzle Weld [Sinw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 600.0 \cdot 9.7573 \cdot 0.49 \cdot 101$
 $= 455074. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 1120.0 \cdot 62.0 \cdot 0.49 \cdot 101$
 $= 5397718. \text{ N}$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 290.5) \cdot (22.0 - 3.0) \cdot 0.7 \cdot 101 \\ &= 1225835. \text{ N} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 600.0 \cdot 62.0 \cdot 0.74 \cdot 101 \\ &= 4366958. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 600.0 \cdot (63.0 - 3.0) \cdot 0.74 \cdot 101 \\ &= 4227242. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (5397718 + 1225835) = 6623553 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (652950 + 4366958 + 4227242 + 455074) = 9702223 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (5397718 + 4227242 + 455074) = 10080033 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 6623553 N	, must exceed W = 3359325 N	or W1 = 3309551 N
Path 2-2 = 9702223 N	, must exceed W = 3359325 N	or W2 = 398090 N
Path 3-3 = 10080033N	, must exceed W = 3359325 N	or W3 = 3678695 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 24.3398 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 2563.0000 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 3.806 %

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Input, Nozzle Desc: K480

From: 30

Pressure for Reinforcement Calculations	P	6.4164	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Hemispherical Head	D	3722.00	mm
Head Finished (Minimum) Thickness	t	63.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1500.0000	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	88.92	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		80.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

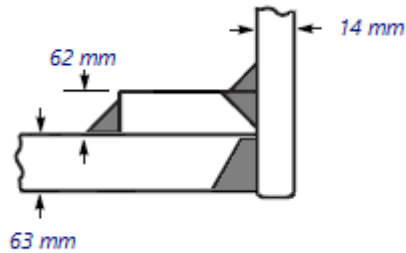
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	190.5000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	62.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	101.03	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	244.0000	mm
Thickness of Pad	te	62.0000	mm
Weld leg size between Pad and Shell	Wp	48.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	62.0000	mm
Reinforcing Pad Width		68.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K480

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	80.000 mm.
Actual Thickness Used in Calculation	14.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 6.42 \cdot 1864.0 / (2 \cdot 101 \cdot 1.0 - 0.2 \cdot 6.42)$
 $= 59.5707 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.42 \cdot 43.0 / (89 \cdot 1.0 - 0.6 \cdot 6.42)$
 $= 3.2433 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.4752 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	283.0802 mm
Parallel to Vessel Wall	Rn+tn+t	141.5401 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		89.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101.0)$
 $= 0.880$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101.0)$
 $= 0.880$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 101.0 / 101.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.88, 1.0)$
 $= 0.880$

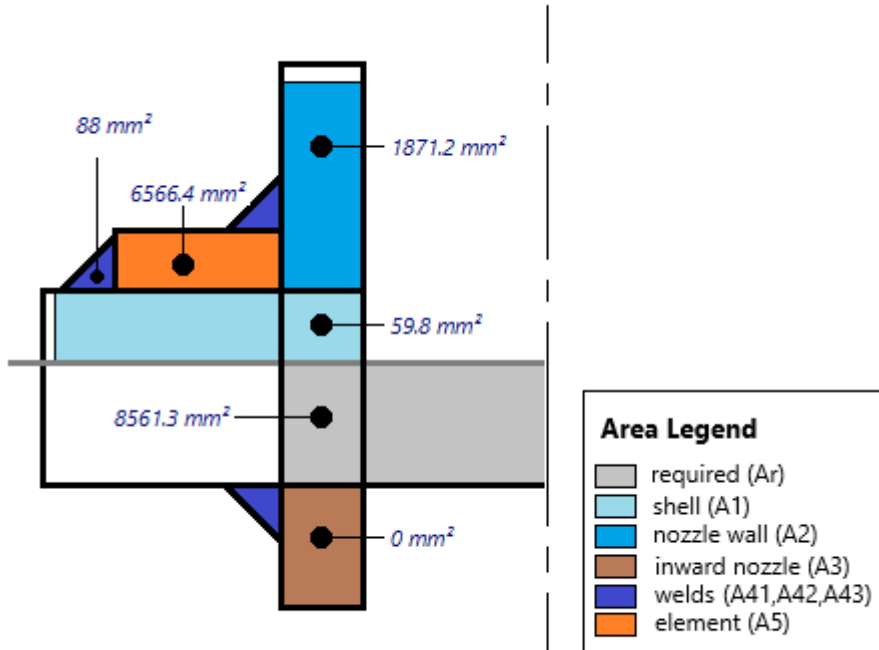


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	8561.308	440.744	NA
Area in Shell	A1	59.830	7507.019	NA
Area in Nozzle Wall	A2	1871.181	2538.924	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	88.015	88.015	NA
Area in Element	A5	6566.403	6566.403	NA
TOTAL AREA AVAILABLE	Atot	8585.429	16700.359	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 37.56 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	213.5207	62.0000 mm
Based on given Pad Diameter:	244.0000	61.7722 mm
Based on the Estimated Diameter Limit:	281.4927	62.7122 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (141.0802 \cdot 59.5707 \cdot 1.0 + 2 \cdot 11.0 \cdot 59.5707 \cdot 1.0 \cdot (1 - 0.88))$
 $= 8561.308 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 142.0(1.0 \cdot 60.0 - 1.0 \cdot 59.571) - 2 \cdot 11.0$
 $(1.0 \cdot 60.0 - 1.0 \cdot 59.5707) \cdot (1 - 0.88)$
 $= 59.830 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$
 $= (2 \cdot 89.5) \cdot (11.0 - 3.24) \cdot 0.8802 / \sin(40.8)$

= 1871.181 mm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10.0^2 \cdot 0.88 + (0.0)^2 \cdot 0.88 + 0.0^2 \cdot 1.0$
 = 88.015 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (Nozzle\ OD)) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(283.0802 - 177.1705) \cdot 62.0 \cdot 1.0$
 = 6566.403 mm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 14.0, tr = 2.417, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.22$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 14.0, tr = 2.417, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.22$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 62.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.726$, Temp. Reduction = 15 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	6 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 14.0, tr = 2.417, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.22$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	:	-104 °C
Gov. MDMT of the Reinforcement Pad	:	6 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	6 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.42/10.20 = 0.629$

Weld Size Calculations, Description: K480

Intermediate Calc. for nozzle/shell Welds	Tmin	11.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	33.9360 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$$

$$= \max(0, (8561.3076 - 59.8301 + 2 \cdot 11.0 \cdot 0.8802 \cdot (1.0 \cdot 60.0 - 59.5707)) \cdot 101)$$

$$= 859650.38 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (1871.1805 + 6566.4028 + 88.0155 - 0.0 \cdot 0.88) \cdot 101$$

$$= 861247.38 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (1871.1805 + 0.0 + 88.0155 + (1161.8043)) \cdot 101$$

$$= 315280.31 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (1871.1805 + 0.0 + 88.0155 + 6566.4028 + (1161.8043)) \cdot 101$$

$$= 978611.69 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 177.1705 \cdot 10.0 \cdot 0.49 \cdot 89$$

$$= 121247. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 244.0 \cdot 48.0 \cdot 0.49 \cdot 101$$

$$= 910647. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 79.5627) \cdot (14.0 - 3.0) \cdot 0.7 \cdot 89$$

$$= 171124. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 177.1705 \cdot 62.0 \cdot 0.74 \cdot 101$$

$$= 1289845. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 177.1705 \cdot (63.0 - 3.0) \cdot 0.74 \cdot 101$$

$$= 1248237. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (910647 + 171124) = 1081771 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (121247 + 1289845 + 1248237 + 0) = 2659329 \text{ N}$$

Failure Path [Path 3-3]:

= (Spew + Tngw + Sinw)
= (910647 + 1248237 + 0) = 2158884 N

Summary of Failure Path Calculations:

Path 1-1 = 1081771 N , must exceed W = 859650 N or W1 = 861247 N
Path 2-2 = 2659329 N , must exceed W = 859650 N or W2 = 315280 N
Path 3-3 = 2158884 N , must exceed W = 859650 N or W3 = 978611 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K480

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	80.000	mm.
Actual Thickness Used in Calculation	14.000	mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $6.42 \cdot 1864.0 / (2 \cdot 101 \cdot 1.0 - 0.2 \cdot 6.42)$
= 59.5707 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.42 \cdot 43.0 / (89 \cdot 1.0 - 0.6 \cdot 6.42)$
= 3.2433 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4752 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	228.0000	mm
Parallel to Vessel Wall	Rn+tn+t	114.0000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		89.5000	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

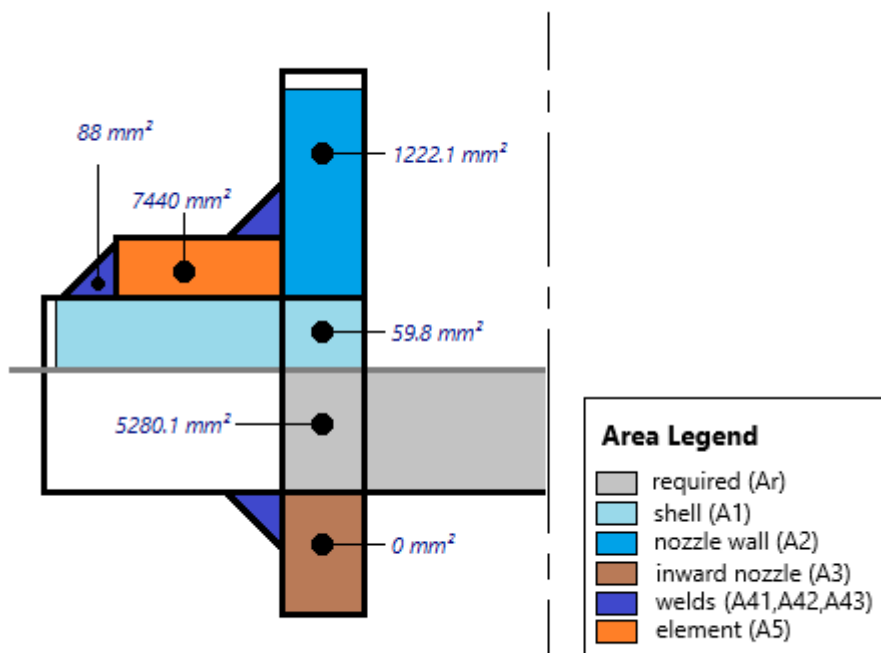


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	5280.143	271.827	NA
Area in Shell	A1	59.830	7507.017	NA
Area in Nozzle Wall	A2	1222.057	1658.157	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	88.015	88.015	NA
Area in Element	A5	7440.000	7440.000	NA
TOTAL AREA AVAILABLE	Atot	8809.902	16693.189	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	171.0684	62.0000 mm
Based on given Pad Diameter:	244.0000	32.5853 mm
Based on the Estimated Diameter Limit:	226.4125	33.0222 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (86.0 \cdot 59.5707 \cdot 1.0 + 2 \cdot 11.0 \cdot 59.5707 \cdot 1.0 \cdot (1 - 0.88))$
 $= 5280.143 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 142.0(1.0 \cdot 60.0 - 1.0 \cdot 59.571) - 2 \cdot 11.0$
 $(1.0 \cdot 60.0 - 1.0 \cdot 59.5707) \cdot (1 - 0.88)$
 $= 59.830 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 89.5) \cdot (11.0 - 3.24) \cdot 0.8802$

$$= 1222.057 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i\text{-can}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10.0^2 \cdot 0.88 + (0.0)^2 \cdot 0.88 + 0.0^2 \cdot 1.0$$

$$= 88.015 \text{ mm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$$

$$= (228.0 - 108.0) \cdot 62.0 \cdot 1.0$$

$$= 7440.000 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 6.2433 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb_1 = 62.5707 \text{ mm}$
Wall Thickness	$tb_1 = \max(trb_1, tr_{16b}) = 62.5707 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb_2 = 3.9542 \text{ mm}$
Wall Thickness	$tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 8.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[8.258, \max(62.5707, 4.5)]$$

$$= 8.2578 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(6.2433, 8.2578)$$

$$= 8.2578 \text{ mm}$$

Available Nozzle Neck Thickness = 14.0000 mm -> OK

Weld Size Calculations, Description: K480

Intermediate Calc. for nozzle/shell Welds	$T_{min} = 11.0000 \text{ mm}$
Intermediate Calc. for pad/shell Welds	$T_{minPad} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code } 7.0700 = 0.7 \cdot W_o \text{ mm}$	
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad} = 33.9360 = 0.7 \cdot W_p \text{ mm}$	

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - tr)) \cdot S_v)$$

$$= \max(0, (5280.1431 - 59.8301 + 2 \cdot 11.0 \cdot 0.8802 \cdot (1.0 \cdot 60.0 - 59.5707)) \cdot 101)$$

$$= 528190.50 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_5 + A_4 - (W_i\text{-Can}/0.707)^2 \cdot fr_2) \cdot S_v$$

$$= (1222.0575 + 7440.0 + 88.0155 - 0.0 \cdot 0.88) \cdot 101$$

$$= 883923.44 \text{ N}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (1222.0575 + 0.0 + 88.0155 + (1161.8043)) \cdot 101$$

$$= 249706.53 \text{ N}$$

Weld Load [W3]:

$$= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$$

$$= (1222.0575 + 0.0 + 88.0155 + 7440.0 + (1161.8043)) \cdot 101$$

$$= 1001287.69 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 108.0 \cdot 10.0 \cdot 0.49 \cdot 89 \\ &= 73910. \text{ N} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.1416/2.0) \cdot 244.0 \cdot 48.0 \cdot 0.49 \cdot 101 \\ &= 910647. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 48.5) \cdot (14.0 - 3.0) \cdot 0.7 \cdot 89 \\ &= 104314. \text{ N} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 108.0 \cdot 62.0 \cdot 0.74 \cdot 101 \\ &= 786267. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 108.0 \cdot (63.0 - 3.0) \cdot 0.74 \cdot 101 \\ &= 760904. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (910647 + 104314) = 1014961 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (73910 + 786267 + 760904 + 0) = 1621080 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (910647 + 760904 + 0) = 1671551 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1014961 N	, must exceed W = 528190 N	or W1 = 883923 N
Path 2-2 = 1621080 N	, must exceed W = 528190 N	or W2 = 249706 N
Path 3-3 = 1671550 N	, must exceed W = 528190 N	or W3 = 1001287 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 77.5921 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 371.4802 mm

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Input, Nozzle Desc: K150

From: 40

Pressure for Reinforcement Calculations	P	6.4164	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3660.00	mm
Design Length of Section	L	4215.6665	mm
Shell Finished (Minimum) Thickness	t	125.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		100.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.02	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

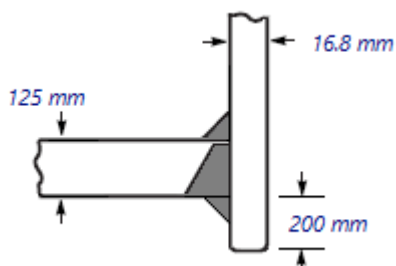
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	124.0000	mm
Inside Projection	h	200.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K150

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 50.800 mm.
Actual Thickness Used in Calculation 16.764 mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.42 \cdot 1833.0 / (101 \cdot 1.0 - 0.6 \cdot 6.42)$
= 121.0273 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.42 \cdot 28.4 / (101 \cdot 1.0 - 0.6 \cdot 6.42)$
= 1.8753 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4974 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	328.3280	mm
Parallel to Vessel Wall	Rn+tn+t	164.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm
Normal to Vessel Wall, Inward		26.9100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K150.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 4.8753	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 124.0273	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 124.0273	mm
Wall Thickness, shell/head, external pressure	trb2	= 4.8776	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.8776	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.8, max(124.0273, 4.8776)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(4.8753, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C



Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $6.42/10.20 = 0.629$

Weld Size Calculations, Description: K150

Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4858 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 624.9999 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 164.16 mm



Nozzle Layout or placement warning. Area for reinforcement may have been used that might not actually exist in this situation. Please check the nozzle geometry and placement carefully.

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Input, Nozzle Desc: K650

From: 40

Pressure for Reinforcement Calculations	P	6.4164	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3660.00	mm
Design Length of Section	L	4215.6665	mm
Shell Finished (Minimum) Thickness	t	125.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2000.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.02	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

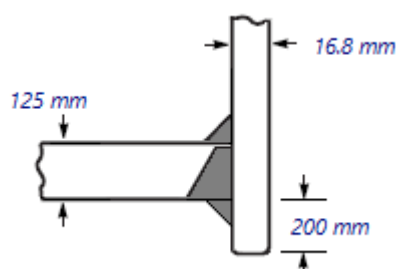
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	124.0000	mm
Inside Projection	h	200.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K650

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 50.800 mm.
Actual Thickness Used in Calculation 16.764 mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.42 \cdot 1833.0 / (101 \cdot 1.0 - 0.6 \cdot 6.42)$
= 121.0273 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.42 \cdot 28.4 / (101 \cdot 1.0 - 0.6 \cdot 6.42)$
= 1.8753 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4974 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	328.3280	mm
Parallel to Vessel Wall	Rn+tn+t	164.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm
Normal to Vessel Wall, Inward		26.9100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K650.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 4.8753	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 124.0273	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 124.0273	mm
Wall Thickness, shell/head, external pressure	trb2	= 4.8776	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.8776	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.8, max(124.0273, 4.8776)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(4.8753, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C



Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $6.42/10.20 = 0.629$

Weld Size Calculations, Description: K650

Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4858 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 624.9999 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 164.16 mm

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Input, Nozzle Desc: K350

From: 40

Pressure for Reinforcement Calculations	P	6.4164	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	+IT
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Cylindrical Shell	D	3660.00	mm
Design Length of Section	L	4215.6665	mm
Shell Finished (Minimum) Thickness	t	125.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2500.00	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105	+IT
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	101.02	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		50.8000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm

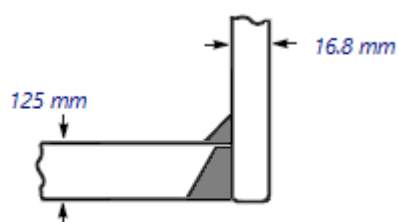
Flange Material		SA-105	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	124.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		600
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: K350

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 50.800 mm.
Actual Thickness Used in Calculation 16.764 mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.42 \cdot 1833.0 / (101 \cdot 1.0 - 0.6 \cdot 6.42)$
= 121.0273 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.42 \cdot 28.4 / (101 \cdot 1.0 - 0.6 \cdot 6.42)$
= 1.8753 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4974 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	328.3280	mm
Parallel to Vessel Wall	Rn+tn+t	164.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	34.4100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K350.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 4.8753	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 124.0273	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 124.0273	mm
Wall Thickness, shell/head, external pressure	trb2	= 4.8776	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.8776	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(124.0273, 4.8776)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(4.8753, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $6.42/10.20 = 0.629$

Weld Size Calculations, Description: K350

Intermediate Calc. for nozzle/shell Welds Tmin 13.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.4858 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 425.4857 mm

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Input, Nozzle Desc: M1600

From: 50

Pressure for Reinforcement Calculations	P	6.4164	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Hemispherical Head	D	3722.00	mm
Head Finished (Minimum) Thickness	t	63.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	101.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		600.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	22.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

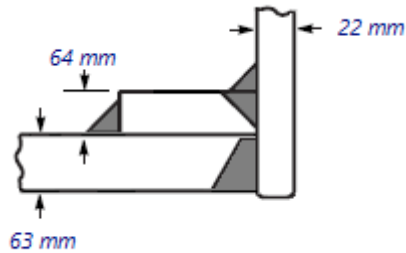
Outside Projection	ho	450.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	62.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	101.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1200.0000	mm
Thickness of Pad	te	64.0000	mm
Weld leg size between Pad and Shell	Wp	56.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	60.0000	mm
Reinforcing Pad Width		300.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		600
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: M1600

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	600.000 mm.
Actual Thickness Used in Calculation	22.000 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 6.42 \cdot 1864.0 / (2 \cdot 101 \cdot 1.0 - 0.2 \cdot 6.42)$
 $= 59.5707 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.42 \cdot 300.0 / (101 \cdot 1.0 + 0.4 \cdot 6.42)$
 $= 18.5862 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.8493 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1124.0001 mm
Parallel to Vessel Wall, opening length	d	562.0001 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		111.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 101.0 / 101.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 101.0 / 101.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 101.0 / 101.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1., 1.)$
 $= 1.000$

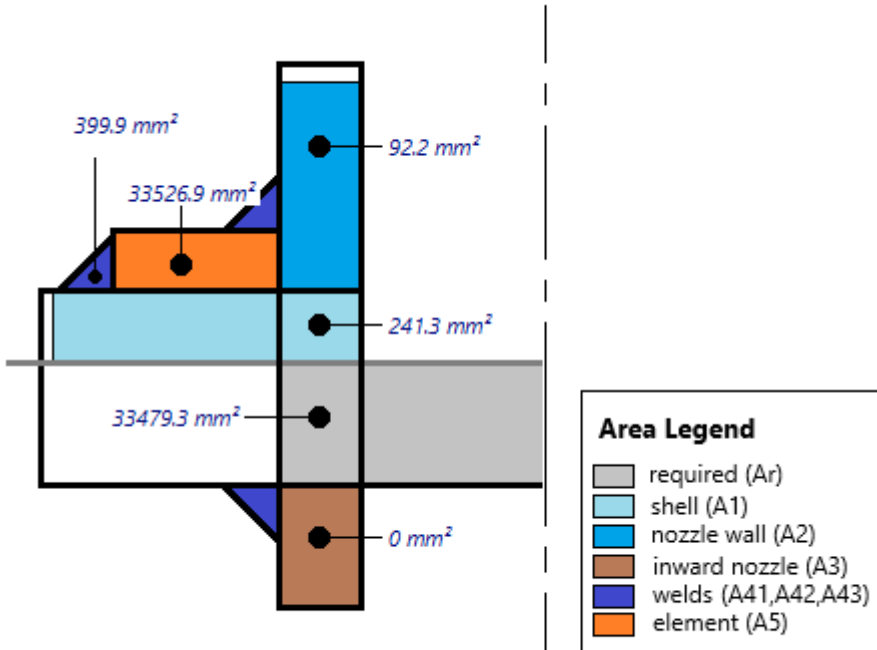


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	33479.348	1723.548	NA
Area in Shell	A1	241.268	30272.410	NA
Area in Nozzle Wall	A2	92.245	3823.552	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	399.891	399.891	NA
Area in Element	A5	33526.852	33526.852	NA
TOTAL AREA AVAILABLE	Atot	34260.254	68022.703	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1111.7952 64.0000 mm
 Based on given Pad Diameter: 1200.0000 62.5093 mm
 Based on the Estimated Diameter Limit: 1122.4126 62.6993 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (562.0 \cdot 59.5707 \cdot 1.0 + 2 \cdot 19.0 \cdot 59.5707 \cdot 1.0 \cdot (1 - 1.))$
 $= 33479.348 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 562.0(1.0 \cdot 60.0 - 1.0 \cdot 59.571) - 2 \cdot 19.0$
 $(1.0 \cdot 60.0 - 1.0 \cdot 59.5707) \cdot (1 - 1.)$
 $= 241.268 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 111.5) \cdot (19.0 - 18.59) \cdot 0.9997$

= 92.245 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
= $20.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 0.0^2 \cdot 1.$
= 399.891 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (Nozzle\ OD)) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
= $(1124.0 - 600.0) \cdot 64.0 \cdot 0.9997$
= 33526.852 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 21.5862$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
= 21.5862 mm

Available Nozzle Neck Thickness = 22.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 22.0$, $t_r = 13.694$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.721$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	15 °C
Min Metal Temp. at Required thickness (UCS 66.1)	0 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 22.0$, $t_r = 13.694$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.721$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-20 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 63.0$, $t_r = 43.537$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.726$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	7 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 22.0$, $t_r = 13.694$, $c = 3.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.721$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-20 °C

Gov. MDMT of the Nozzle	:	0 °C
Gov. MDMT of the Reinforcement Pad	:	7 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	7 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :



Design Pressure/Ambient Rating = 6.42/10.20 = 0.629

Weld Size Calculations, Description: M1600

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	13.3000 = 0.7 * tmin.	14.1400 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	39.5920 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (33479.3477 - 241.2678 + 2 \cdot 19.0 \cdot 0.9997 \cdot (1.0 \cdot 60.0 - 59.5707)) \cdot 101)$
 $= 3359325.00 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (92.2454 + 33526.8516 + 399.8908 - 0.0 \cdot 1.) \cdot 101$
 $= 3436563.75 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (92.2454 + 0.0 + 399.8908 + (2279.3774)) \cdot 101$
 $= 279975.53 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (92.2454 + 0.0 + 399.8908 + 33526.8516 + (2279.3774)) \cdot 101$
 $= 3666824.25 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 600.0 \cdot 20.0 \cdot 0.49 \cdot 101$
 $= 932785. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 1200.0 \cdot 56.0 \cdot 0.49 \cdot 101$
 $= 5223598. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 290.5) \cdot (22.0 - 3.0) \cdot 0.7 \cdot 101$
 $= 1225835. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 600.0 \cdot 60.0 \cdot 0.74 \cdot 101$
 $= 4226088. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 600.0 \cdot (63.0 - 3.0) \cdot 0.74 \cdot 101$
 $= 4227242. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (5223598 + 1225835) = 6449433 \text{ N}$



Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$
$$= (932785 + 4226089 + 4227242 + 0) = 9386116 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$
$$= (5223598 + 4227242 + 0) = 9450840 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 6449433 N	, must exceed W = 3359325 N	or W1 = 3436563 N
Path 2-2 = 9386116 N	, must exceed W = 3359325 N	or W2 = 279975 N
Path 3-3 = 9450840 N	, must exceed W = 3359325 N	or W3 = 3666824 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 24.3398 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 537.3398 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{\text{nom}} / R_f \cdot (1 - R_f / R_o)$) 3.806 %

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Input, Nozzle Desc: K250

From: 50

Pressure for Reinforcement Calculations	P	6.4164	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Hemispherical Head	D	3722.00	mm
Head Finished (Minimum) Thickness	t	63.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1500.0000	mm
-------------------------------	----	-----------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	88.90	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		50.0000	mm

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.8000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

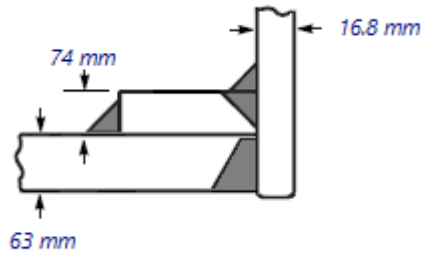
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	800.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	62.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	101.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	184.3280	mm
Thickness of Pad	te	74.0000	mm
Weld leg size between Pad and Shell	Wp	56.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	74.0000	mm
Reinforcing Pad Width		50.3640	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K250

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	50.000 mm.
Actual Thickness Used in Calculation	16.800 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 6.42 \cdot 1864.0 / (2 \cdot 101 \cdot 1.0 - 0.2 \cdot 6.42)$
 $= 59.5707 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 6.42 \cdot 28.0 / (89 \cdot 1.0 - 0.6 \cdot 6.42)$
 $= 2.1124 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.7478 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	239.3921 mm
Parallel to Vessel Wall	Rn+tn+t	119.6960 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		108.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101.0)$
 $= 0.880$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101.0)$
 $= 0.880$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 101.0 / 101.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.88, 1.)$
 $= 0.880$

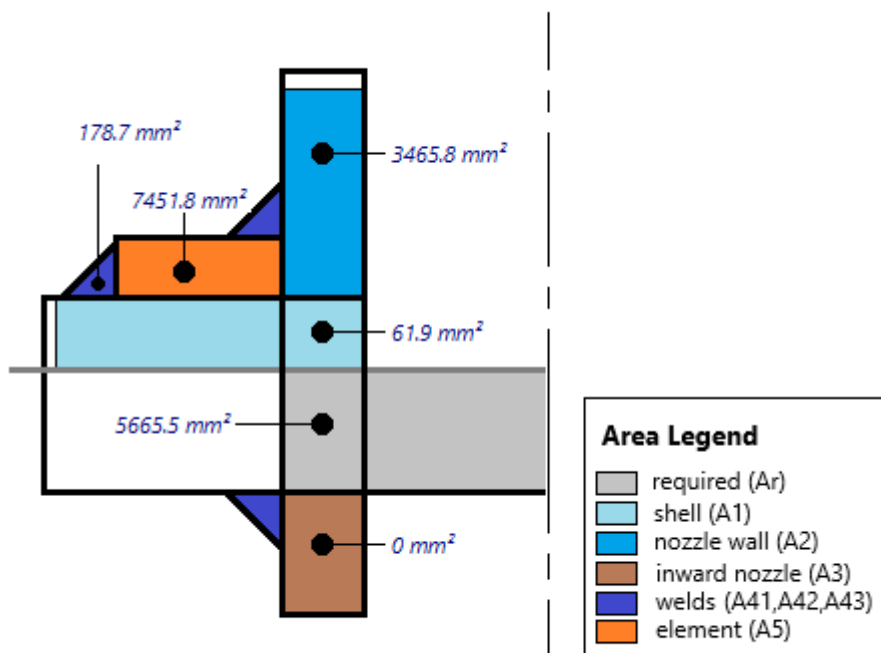


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	5665.484	291.664	NA
Area in Shell	A1	61.944	7772.227	NA
Area in Nozzle Wall	A2	3465.769	3870.401	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	178.678	178.678	NA
Area in Element	A5	7451.837	7451.837	NA
TOTAL AREA AVAILABLE	Atot	11158.228	19273.141	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 37.59 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	110.0815	74.0000 mm
Based on given Pad Diameter:	184.3280	19.4547 mm
Based on the Estimated Diameter Limit:	237.8046	19.4461 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (91.7921 \cdot 59.5707 \cdot 1.0 + 2 \cdot 13.8 \cdot 59.5707 \cdot 1.0 \cdot (1 - 0.88))$$

$$= 5665.484 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 147.6(1.0 \cdot 60.0 - 1.0 \cdot 59.571) - 2 \cdot 13.8$$

$$(1.0 \cdot 60.0 - 1.0 \cdot 59.5707) \cdot (1 - 0.88)$$

$$= 61.944 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$$

$$= (2 \cdot 108.5) \cdot (13.8 - 2.11) \cdot 0.88 / \sin(40.1)$$



= 3465.769 mm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (100.0) • 0.88 + (0.0) • 0.88 + 90.7066² • 1.
= 178.678 mm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (237.7604 - 137.0324) • 74.0 • 0.9997
= 7451.837 mm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 16.8, tr = 1.574, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.114, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 16.8, tr = 1.574, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.114, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 63.0, tr = 43.537, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.726, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	7 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 1.574, c = 3.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.114, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	:	-104 °C
Gov. MDMT of the Reinforcement Pad	:	7 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	7 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 6.42/10.20 = 0.629

Weld Size Calculations, Description: K250

Intermediate Calc. for nozzle/shell Welds	Tmin	13.8000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	39.5920 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t-tr)) \cdot S_v)$$

$$= \max(0, (5665.4839 - 61.9438 + 2 \cdot 13.8 \cdot 0.88 \cdot (1.0 \cdot 60.0 - 59.5707)) \cdot 101)$$

$$= 567117.19 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_r2) \cdot S_v$$

$$= (3465.7688 + 7451.8369 + 178.6776 - 0.0 \cdot 0.88) \cdot 101$$

$$= 1120935.38 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$$

$$= (3465.7688 + 0.0 + 87.9958 + (1457.21)) \cdot 101$$

$$= 506203.56 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (3465.7688 + 0.0 + 178.6776 + 7451.8369 + (1457.21)) \cdot 101$$

$$= 1268141.25 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 137.0324 \cdot 10.0 \cdot 0.49 \cdot 89$$

$$= 93757. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 184.328 \cdot 56.0 \cdot 0.49 \cdot 101$$

$$= 802379. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 57.2061) \cdot (16.8 - 3.0) \cdot 0.7 \cdot 89$$

$$= 154324. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 137.0324 \cdot 74.0 \cdot 0.74 \cdot 101$$

$$= 1190395. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 137.0324 \cdot (63.0 - 3.0) \cdot 0.74 \cdot 101$$

$$= 965449. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (802379 + 154324) = 956704 \text{ N}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (93757 + 1190395 + 965449 + 0) = 2249601 \text{ N}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$



= (802379 + 965449 + 0) = 1767828 N

Summary of Failure Path Calculations:

Path 1-1 = 956703 N , must exceed W = 567117 N or W1 = 1120935 N
Path 2-2 = 2249600 N , must exceed W = 567117 N or W2 = 506203 N
Path 3-3 = 1767828 N , must exceed W = 567117 N or W3 = 1268141 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K250

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 50.000 mm.
Actual Thickness Used in Calculation 16.800 mm

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $6.42 \cdot 1864.0 / (2 \cdot 101 \cdot 1.0 - 0.2 \cdot 6.42)$
= 59.5707 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $6.42 \cdot 28.0 / (89 \cdot 1.0 - 0.6 \cdot 6.42)$
= 2.1124 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.7478 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	203.6000 mm
Parallel to Vessel Wall	Rn+tn+t	101.8000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		108.5000 mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K250.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 5.1124 mm
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 62.5707 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 62.5707 mm
Wall Thickness, shell/head, external pressure	trb2	= 3.9542 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm
Wall Thickness per table UG-45	tb3	= 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[7.8, max(62.5707, 4.5)]
= 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:



= max(ta, tb)
= max(5.1124, 7.8)
= 7.8000 mm

Available Nozzle Neck Thickness = 16.8000 mm -> OK

Weld Size Calculations, Description: K250

Intermediate Calc. for nozzle/shell Welds	Tmin	13.8000	mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	39.5920 = 0.7 * Wp mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 59.3118 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 962.7000 mm

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Input, Nozzle Desc: K580

From: 50

Pressure for Reinforcement Calculations	P	6.4164	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Hemispherical Head	D	3722.00	mm
Head Finished (Minimum) Thickness	t	63.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1500.0000	mm
-------------------------------	----	-----------	----

User Entered Minimum Design Metal Temperature		-12.00	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	88.90	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		80.0000	mm

Size and Thickness Basis		Minimum	
Nominal Thickness		160	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

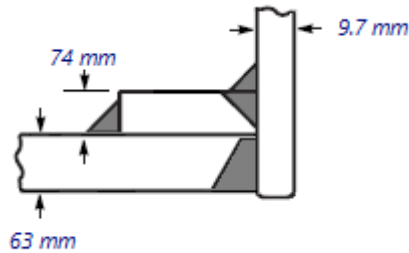
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	1200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	62.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	101.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	288.9000	mm
Thickness of Pad	te	74.0000	mm
Weld leg size between Pad and Shell	Wp	56.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	74.0000	mm
Reinforcing Pad Width		100.0000	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: K580

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	88.900 mm.
Actual Thickness Used in Calculation	9.735 mm

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 6.42 \cdot 1864.0 / (2 \cdot 101 \cdot 1.0 - 0.2 \cdot 6.42)$
 $= 59.5707 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.42 \cdot 44.45 / (89 \cdot 1.0 + 0.4 \cdot 6.42)$
 $= 3.1182 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.8297 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	257.1709 mm
Parallel to Vessel Wall	Rn+tn+t	128.5854 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		90.8364 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101.0)$
 $= 0.880$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101.0)$
 $= 0.880$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 101.0 / 101.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.88, 1.)$
 $= 0.880$

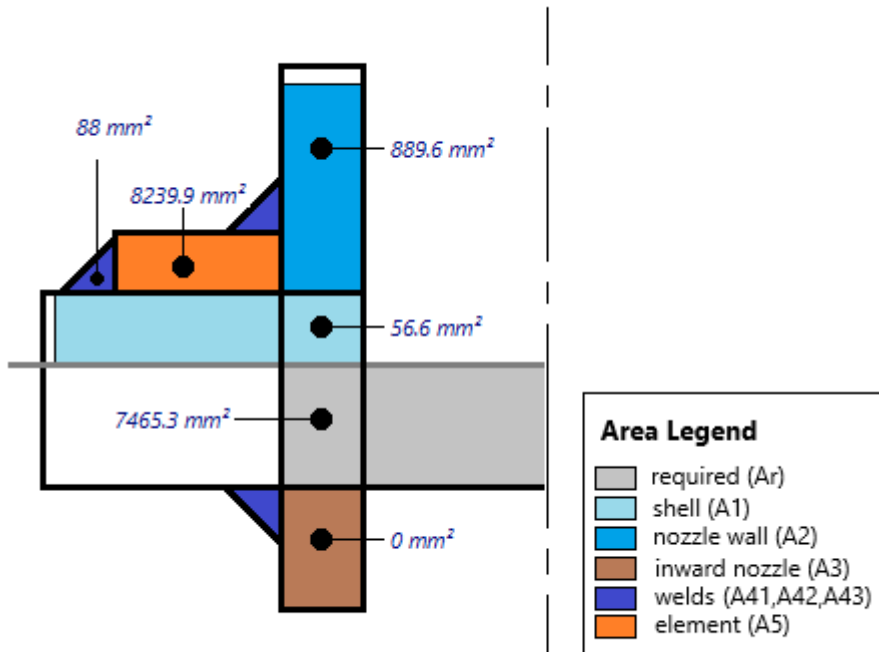


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	7465.316	384.322	NA
Area in Shell	A1	56.605	7102.417	NA
Area in Nozzle Wall	A2	889.575	1452.502	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	87.996	87.996	NA
Area in Element	A5	8239.919	8239.919	NA
TOTAL AREA AVAILABLE	Atot	9274.096	16882.834	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 37.57 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	175.8310	74.0000 mm
Based on given Pad Diameter:	288.9000	57.7560 mm
Based on the Estimated Diameter Limit:	255.5834	58.5910 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (123.7018 \cdot 59.5707 \cdot 1.0 + 2 \cdot 6.7346 \cdot 59.5707 \cdot 1.0 \cdot (1 - 0.88))$
 $= 7465.316 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 133.469(1.0 \cdot 60.0 - 1.0 \cdot 59.571) - 2 \cdot 6.735$
 $(1.0 \cdot 60.0 - 1.0 \cdot 59.5707) \cdot (1 - 0.88)$
 $= 56.605 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$
 $= (2 \cdot 90.84) \cdot (6.73 - 3.12) \cdot 0.88 / \sin(40.5)$

= 889.575 mm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i - can / 0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10.0^2 \cdot 0.88 + (0.0)^2 \cdot 0.88 + 0.0^2 \cdot 1.$
 $= 87.996 \text{ mm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr4$
 $= (257.1709 - 145.7902) \cdot 74.0 \cdot 0.9997$
 $= 8239.919 \text{ mm}^2$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 9.735$, $t_r = 2.366$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.351$, Temp. Reduction = 77 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 9.735$, $t_r = 2.366$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.351$, Temp. Reduction = 77 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 63.0$, $t_r = 43.537$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.726$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	7 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 9.735$, $t_r = 2.366$, $c = 3.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.351$, Temp. Reduction = 77 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Gov. MDMT of the Nozzle	:	-48 °C
Gov. MDMT of the Reinforcement Pad	:	7 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	7 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.42 / 10.20 = 0.629$

Weld Size Calculations, Description: K580

Intermediate Calc. for nozzle/shell Welds	Tmin	6.7346	mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000	mm



Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.7142 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 * t_{minPad}$	$39.5920 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$\begin{aligned} &= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t-tr)) \cdot S_v) \\ &= \max(0, (7465.3164 - 56.6055 + 2 \cdot 6.7346 \cdot 0.88 \cdot (1.0 \cdot 60.0 - 59.5707)) \cdot 101) \\ &= 748934.50 \text{ N} \end{aligned}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$\begin{aligned} &= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_r2) \cdot S_v \\ &= (889.5751 + 8239.9189 + 87.9958 - 0.0 \cdot 0.88) \cdot 101 \\ &= 931141.44 \text{ N} \end{aligned}$$

Weld Load [W2]:

$$\begin{aligned} &= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v \\ &= (889.5751 + 0.0 + 87.9958 + (711.1344)) \cdot 101 \\ &= 170591.30 \text{ N} \end{aligned}$$

Weld Load [W3]:

$$\begin{aligned} &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S \\ &= (889.5751 + 0.0 + 87.9958 + 8239.9189 + (711.1344)) \cdot 101 \\ &= 1002979.50 \text{ N} \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 145.7902 \cdot 10.0 \cdot 0.49 \cdot 89 \\ &= 99749. \text{ N} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW} \\ &= (3.1416/2.0) \cdot 288.9 \cdot 56.0 \cdot 0.49 \cdot 101 \\ &= 1257581. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 67.373) \cdot (9.7346 - 3.0) \cdot 0.7 \cdot 89 \\ &= 88697. \text{ N} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg} \\ &= (3.1416/2) \cdot 145.7902 \cdot 74.0 \cdot 0.74 \cdot 101 \\ &= 1266474. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 145.7902 \cdot (63.0 - 3.0) \cdot 0.74 \cdot 101 \\ &= 1027151. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (1257581 + 88697) = 1346278 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw}) \\ &= (99749 + 1266474 + 1027151 + 0) = 2393374 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$



$$= (1257581 + 1027151 + 0) = 2284732 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 1346277 N , must exceed W = 748934 N or W1 = 931141 N
Path 2-2 = 2393373 N , must exceed W = 748934 N or W2 = 170591 N
Path 3-3 = 2284731 N , must exceed W = 748934 N or W3 = 1002979 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: K580

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	88.900	mm.
Actual Thickness Used in Calculation	9.735	mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
= $P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
= $6.42 \cdot 1864.0 / (2 \cdot 101 \cdot 1.0 - 0.2 \cdot 6.42)$
= 59.5707 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $6.42 \cdot 44.45 / (89 \cdot 1.0 + 0.4 \cdot 6.42)$
= 3.1182 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.8297 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	208.9000	mm
Parallel to Vessel Wall	Rn+tn+t	104.4500	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		90.8364	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

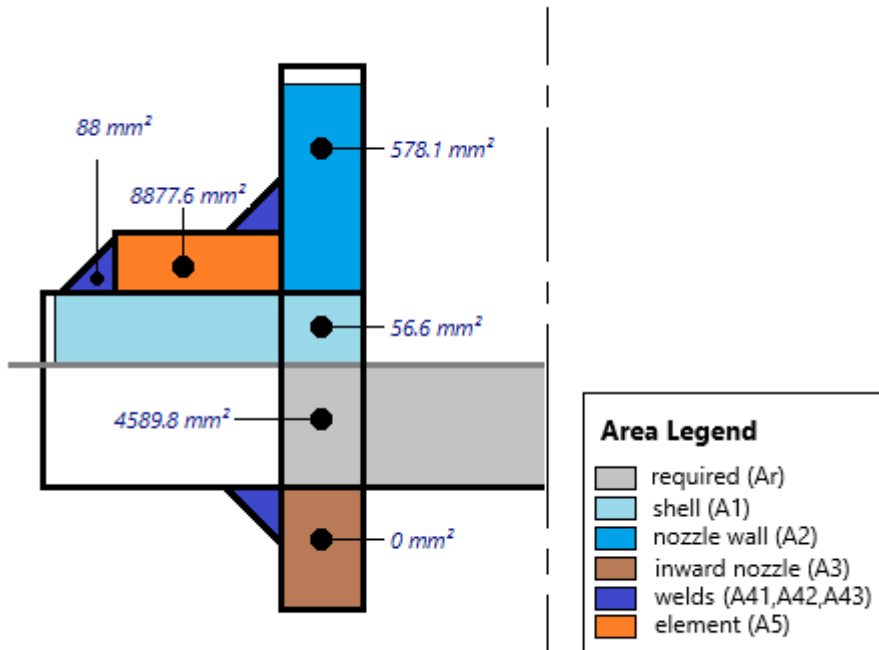


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	4589.788	236.287	NA
Area in Shell	A1	56.605	7102.417	NA
Area in Nozzle Wall	A2	578.131	943.975	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	87.996	87.996	NA
Area in Element	A5	8877.575	8877.575	NA
TOTAL AREA AVAILABLE	Atot	9600.308	17011.963	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 141.1718 74.0000 mm
 Based on given Pad Diameter: 288.9000 32.2343 mm
 Based on the Estimated Diameter Limit: 207.3125 32.6664 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (75.4309 \cdot 59.5707 \cdot 1.0 + 2 \cdot 6.7346 \cdot 59.5707 \cdot 1.0 \cdot (1 - 0.88))$
 $= 4589.788 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 133.469(1.0 \cdot 60.0 - 1.0 \cdot 59.571) - 2 \cdot 6.735$
 $(1.0 \cdot 60.0 - 1.0 \cdot 59.5707) \cdot (1 - 0.88)$
 $= 56.605 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 90.84) \cdot (6.73 - 3.12) \cdot 0.88$



= 578.131 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i\text{-can}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10.0^2 \cdot 0.88 + (0.0)^2 \cdot 0.88 + 0.0^2 \cdot 1.$
 = 87.996 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$
 = $(208.9 - 88.9) \cdot 74.0 \cdot 0.9997$
 = 8877.575 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 6.1182 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 62.5707 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 62.5707 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.9542 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[7.8, max(62.5707, 4.5)]
 = 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(6.1182, 7.8)
 = 7.8000 mm

Available Nozzle Neck Thickness = 9.7346 mm -> OK

Weld Size Calculations, Description: K580

Intermediate Calc. for nozzle/shell Welds	Tmin	6.7346 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.7142 = $0.7 \cdot t_{min}$	7.0700 = $0.7 \cdot W_o$ mm
Pad Weld	9.5000 = $0.5 \cdot T_{minPad}$	39.5920 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 = $\max(0, (A - A1 + 2 \cdot t_n \cdot fr_1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 = $\max(0, (4589.7881 - 56.6055 + 2 \cdot 6.7346 \cdot 0.88 \cdot (1.0 \cdot 60.0 - 59.5707)) \cdot 101)$
 = 458451.53 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 = $(A2 + A5 + A4 - (W_i\text{-Can}/0.707)^2 \cdot fr_2) \cdot Sv$
 = $(578.1312 + 8877.5752 + 87.9958 - 0.0 \cdot 0.88) \cdot 101$
 = 964095.12 N

Weld Load [W2]:
 = $(A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot Sv$
 = $(578.1312 + 0.0 + 87.9958 + (711.1344)) \cdot 101$
 = 139129.55 N

Weld Load [W3]:
 = $(A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S$
 = $(578.1312 + 0.0 + 87.9958 + 8877.5752 + (711.1344)) \cdot 101$
 = 1035933.19 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw \\ &= (3.1416/2.0) \cdot 88.9 \cdot 10.0 \cdot 0.49 \cdot 89 \\ &= 60825. \text{ N} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.1416/2.0) \cdot 288.9 \cdot 56.0 \cdot 0.49 \cdot 101 \\ &= 1257581. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 41.0827) \cdot (9.7346 - 3.0) \cdot 0.7 \cdot 89 \\ &= 54085. \text{ N} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 88.9 \cdot 74.0 \cdot 0.74 \cdot 101 \\ &= 772271. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 88.9 \cdot (63.0 - 3.0) \cdot 0.74 \cdot 101 \\ &= 626336. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1257581 + 54085) = 1311667 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ &= (60825 + 772271 + 626336 + 0) = 1459432 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ &= (1257581 + 626336 + 0) = 1883918 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 1311666 N	, must exceed W = 458451 N	or W1 = 964095 N
Path 2-2 = 1459432 N	, must exceed W = 458451 N	or W2 = 139129 N
Path 3-3 = 1883917 N	, must exceed W = 458451 N	or W3 = 1035933 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 63.2430 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1366.6312 mm

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Input, Nozzle Desc: N1400

From: 50

Pressure for Reinforcement Calculations	P	6.4164	MPa
Temperature for Internal Pressure	Temp	400	°C
Design External Pressure	Pext	0.10	MPa
Temperature for External Pressure	Tempex	400	°C

Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	101.03	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa

Inside Diameter of Hemispherical Head	D	3722.00	mm
Head Finished (Minimum) Thickness	t	63.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	1500.0000	mm
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User Entered Minimum Design Metal Temperature		-12.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	88.90	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		400.0000	mm

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

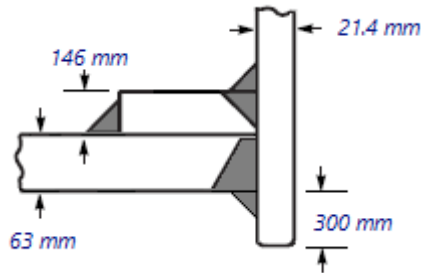
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	62.0000	mm
Inside Projection	h	300.0000	mm
Weld leg size, Inside Element to Shell	Wi	14.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	101.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	723.8500	mm
Thickness of Pad	te	146.0000	mm
Weld leg size between Pad and Shell	Wp	110.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	146.0000	mm
Reinforcing Pad Width		158.7250	mm

Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N1400

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	406.400 mm.
Actual Thickness Used in Calculation	21.412 mm

Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Spherical Head, tr [Int. Press]
 $= P \cdot R / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per UG-27 (d)
 $= 6.42 \cdot 1864.0 / (2 \cdot 101 \cdot 1.0 - 0.2 \cdot 6.42)$
 $= 59.5707 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 6.42 \cdot 203.2 / (89 \cdot 1.0 + 0.4 \cdot 6.42)$
 $= 14.2545 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0528 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	739.1512 mm
Parallel to Vessel Wall, opening length	d	369.5756 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		150.0000 mm
Normal to Vessel Wall, Inward		38.5305 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101.0)$
 $= 0.880$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.9 / 101.0)$
 $= 0.880$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 101.0 / 101.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.88, 1.)$
 $= 0.880$

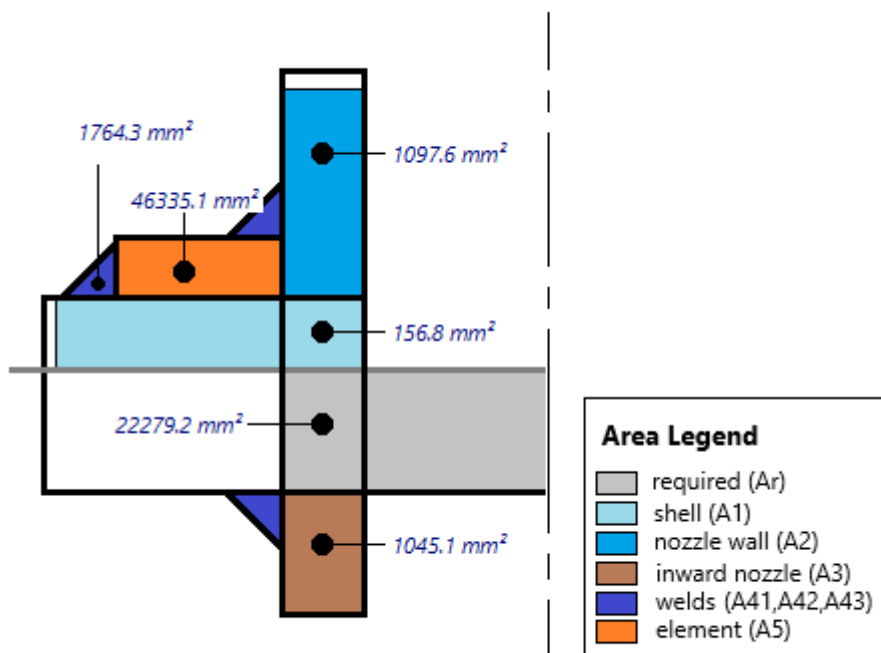


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	22279.207	1146.954	NA
Area in Shell	A1	156.765	19669.625	NA
Area in Nozzle Wall	A2	1097.590	4582.663	NA
Area in Inward Nozzle	A3	1045.108	1045.108	NA
Area in Welds	A41+A42+A43	1764.260	1764.260	NA
Area in Element	A5	46335.051	46335.051	NA
TOTAL AREA AVAILABLE	Atot	50398.773	73396.703	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	531.1976	146.0000 mm
Based on given Pad Diameter:	723.8500	57.3963 mm
Based on the Estimated Diameter Limit:	737.5637	55.0195 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (369.5756 \cdot 59.5707 \cdot 1.0 + 2 \cdot 18.4122 \cdot 59.5707 \cdot 1.0 \cdot (1 - 0.88))$
 $= 22279.207 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 369.576(1.0 \cdot 60.0 - 1.0 \cdot 59.571) - 2 \cdot 18.412$
 $(1.0 \cdot 60.0 - 1.0 \cdot 59.5707) \cdot (1 - 0.88)$
 $= 156.765 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 150.0) \cdot (18.41 - 14.25) \cdot 0.88$

= 1097.590 mm²

Area Available in Inward Nozzle [A3]:

= $2 \cdot t_i \cdot \min(h, Tl, 2.5 \cdot t_i) \cdot fr2$
= $2 \cdot 15.4122 \cdot (38.5305) \cdot 0.88$
= 1045.108 mm²

Area Available in Welds [A41 + A42 + A43]:

= $(Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can/0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$
= $(64.0) \cdot 0.88 + (95.2053) \cdot 0.88 + 1624.6099^2 \cdot 1.$
= 1764.260 mm²

Area Available in Element [A5]:

= $(\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$
= $(723.85 - 406.4) \cdot 146.0 \cdot 0.9997$
= 46335.051 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 17.2545 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 62.5707 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 62.5707 mm
Wall Thickness, shell/head, external pressure	trb2 = 3.9542 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]
= min[11.331, max(62.5707, 4.5)]
= 11.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(17.2545, 11.3312)
= 17.2545 mm

Available Nozzle Neck Thickness = $0.875 \cdot 21.412 = 18.736$ mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 54.0,	Allowable	: 88.9 MPa	Passed
Expansion	: 0.0,	Allowable	: 204.6 MPa	Passed
Occasional	: 30.7,	Allowable	: 118.2 MPa	Passed
Shear	: 12.8,	Allowable	: 62.2 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 18.736, tr = 10.814, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.687$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	10 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-7 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 18.736, tr = 10.814, c = 3.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.687$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 18.736, tr = 10.814, c = 3.0 mm, E* = 1.0



Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.687$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -27 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $tg = 63.0$, $tr = 43.537$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.726$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 22 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 7 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 18.736$, $tr = 10.814$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.687$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -27 °C

Gov. MDMT of the Nozzle : -7 °C
 Gov. MDMT of the Reinforcement Pad : 7 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 7 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -38 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $6.42/10.20 = 0.629$

Weld Size Calculations, Description: N1400

Intermediate Calc. for nozzle/shell Welds	Tmin	18.4122	mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000	mm
Intermediate Calc. for Inward Weld	TminIns	18.4122	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$77.7700 = 0.7 \cdot W_p \text{ mm}$
Inward Weld	$6.0000 = \text{Min per Code}$	$6.8980 = 0.7 \cdot W_i\text{-Can mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (22279.207 - 156.7648 + 2 \cdot 18.4122 \cdot 0.88 \cdot (1.0 \cdot 60.0 - 59.5707)) \cdot 101)$
 $= 2236192.25 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i\text{-Can}/.707)^2 \cdot fr2) \cdot Sv$
 $= (1097.5896 + 46335.0508 + 1764.2603 - 95.2053 \cdot 0.88) \cdot 101$
 $= 4961358.50 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (1097.5896 + 1045.1079 + 171.7724 + (1944.2352)) \cdot 101$
 $= 430210.06 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (1097.5896 + 1045.1079 + 1764.2603 + 46335.0508 + (1944.2352)) \cdot 101$
 $= 5271802.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 406.4 \cdot 10.0 \cdot 0.49 \cdot 89 \\ &= 278057. \text{ N} \end{aligned}$$

Shear, Inward Nozzle Weld [Sinw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\ &= (3.1416/2.0) \cdot 406.4 \cdot 9.7573 \cdot 0.49 \cdot 89 \\ &= 271310. \text{ N} \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW} \\ &= (3.1416/2.0) \cdot 723.85 \cdot 110.0 \cdot 0.49 \cdot 101 \\ &= 6189302. \text{ N} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n \\ &= (3.1416 \cdot 193.9939) \cdot (21.4122 - 3.0) \cdot 0.7 \cdot 89 \\ &= 698243. \text{ N} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg} \\ &= (3.1416/2) \cdot 406.4 \cdot 146.0 \cdot 0.74 \cdot 101 \\ &= 6965344. \text{ N} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng} \\ &= (3.1416/2.0) \cdot 406.4 \cdot (63.0 - 3.0) \cdot 0.74 \cdot 101 \\ &= 2863252. \text{ N} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (6189303 + 698243) = 6887546 \text{ N}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (278057 + 6965345 + 2863252 + 271310) = 10377963 \text{ N} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (6189303 + 2863252 + 271310) = 9323864 \text{ N} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 6887545 N	, must exceed W = 2236192 N	or W1 = 4961358 N
Path 2-2 = 10377963N	, must exceed W = 2236192 N	or W2 = 430210 N
Path 3-3 = 9323864 N	, must exceed W = 2236192 N	or W3 = 5271802 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 6 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 11.1268 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 563.0000 mm

Input Echo, WRC107/537 Item 1, Description: N1400 :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	3.0000 mm
Vessel Diameter	Dv	3722.000 mm
Vessel Thickness	Tv	63.000 mm

Design Temperature	T1	400.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	101.03	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	406.400	mm
Nozzle Thickness	Tn	21.412	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	88.90	MPa
Thickness of Reinforcing Pad	Tpad	146.000	mm
Diameter of Reinforcing Pad	Dpad	723.850	mm
Design Internal Pressure	Dp	6.416	MPa
Include Pressure Thrust		No	

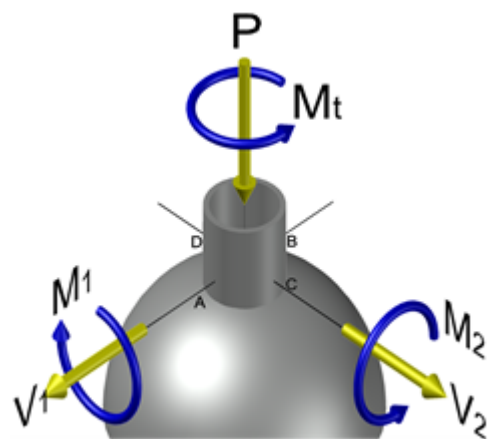
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	36000.0	N
Longitudinal Shear (SUS)	(V1) V1	26000.0	N
Circumferential Shear (SUS)	(Vc) V2	26000.0	N
Circumferential Moment (SUS)	(Mc) M1	32000000.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	32000000.0	N-mm
Torsional Moment (SUS)	Mt	36799944.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 406.4 + 2 \cdot 1.65 \cdot \sqrt{(1894.0 (63.0 - 3.0))}$$

$$= 1518.848 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	36000.0	N
Circumferential Shear	(VC) V2	26000.0	N
Longitudinal Shear	(VL) V1	26000.0	N
Circumferential Moment	(MC) M1	32000000.0	N-mm
Longitudinal Moment	(ML) M2	32000000.0	N-mm
Torsional Moment	MT	36799944.0	N-mm

Dimensionless Param: U = 0.32 TAU = 10.54 RHO = 4.00 (11.19)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 4	0.06127	(A,B,C,D)
M(x) / P	SP 4	0.02540	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 4	0.18480	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 4	0.08284	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 4	0.18480	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 4	0.08284	(A,B,C,D)
N(y) * T / P	SP 4	0.34219	(A,B,C,D)
M(y) / P	SP 4	0.12973	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 4	0.26780	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 4	0.48880	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 4	0.26780	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 4	0.48880	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1
Rad. Bend.	P	-0.1	0.1	-0.1	0.1	-0.1	0.1	-0.1	0.1
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-0.6	0.6	0.6	-0.6
Rad. Memb.	ML	-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-0.6	0.6	0.6	-0.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-1.0	0.4	0.6	-0.3	-1.0	0.4	0.6	-0.3
Tang. Memb.	P	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3	-0.3
Tang. Bend.	P	-0.7	0.7	-0.7	0.7	-0.7	0.7	-0.7	0.7
Tang. Memb.	MC	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Tang. Bend.	MC	0.0	0.0	0.0	0.0	-3.5	3.5	3.5	-3.5
Tang. Memb.	ML	-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Tang. Bend.	ML	-3.5	3.5	3.5	-3.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-4.7	3.5	2.8	-2.8	-4.7	3.5	2.8	-2.8
Shear VC		0.2	0.2	-0.2	-0.2	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Shear MT		0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Tot. Shear		0.9	0.9	0.5	0.5	0.5	0.5	0.9	0.9
Str. Int.		4.9	3.8	2.9	2.9	4.8	3.6	3.2	3.1

Unitless Prm: U = 1.07 TAU = 0.00 (19.16) RHO = 0.00 (3.26)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
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Pm (SUS)	29.3	29.3	29.3	29.3	29.3	29.3	29.3	29.3
Pm+Pl (SUS)	29.1	29.1	29.6	29.6	29.1	29.1	29.6	29.6
Pm+Pl+Q (Total)	28.5	33.0	32.2	29.1	28.3	32.9	32.4	29.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	29.25	101.03	Passed
Pm+Pl (SUS)	29.56	151.54	Passed
Pm+Pl+Q (TOTAL)	33.02	303.08	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	99.7	99.7	99.7	99.7	99.7	99.7	99.7	99.7
Rad.	Pl (SUS)	-2.4	-2.4	1.2	1.2	-2.4	-2.4	1.2	1.2
Rad.	Q (SUS)	-11.7	11.7	7.9	-7.9	-11.7	11.7	7.9	-7.9
Long.	Pm (SUS)	99.7	99.7	99.7	99.7	99.7	99.7	99.7	99.7
Long.	Pl (SUS)	-0.7	-0.7	0.4	0.4	-0.7	-0.7	0.4	0.4
Long.	Q (SUS)	-3.5	3.5	2.4	-2.4	-3.5	3.5	2.4	-2.4
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.4	0.4	-0.4	-0.4	-0.4	-0.4	0.4	0.4
Shear	Q (SUS)	0.7	0.7	0.7	0.7	0.7	0.7	0.7	0.7
Pm (SUS)		99.7	99.7	99.7	99.7	99.7	99.7	99.7	99.7
Pm+Pl (SUS)		99.1	99.1	101.1	101.1	99.1	99.1	101.1	101.1
Pm+Pl+Q (Total)		95.6	109.2	108.9	97.7	95.5	109.0	109.1	98.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	99.74	101.03	Passed
Pm+Pl (SUS)	101.10	151.54	Passed
Pm+Pl+Q (TOTAL)	109.18	303.08	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: One or more nozzles do not meet proper Spacing Requirements!**

Please review all reports carefully!

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